POLITECNICO DI MILANO

Biochip Course

Optical Detection for Biosensing

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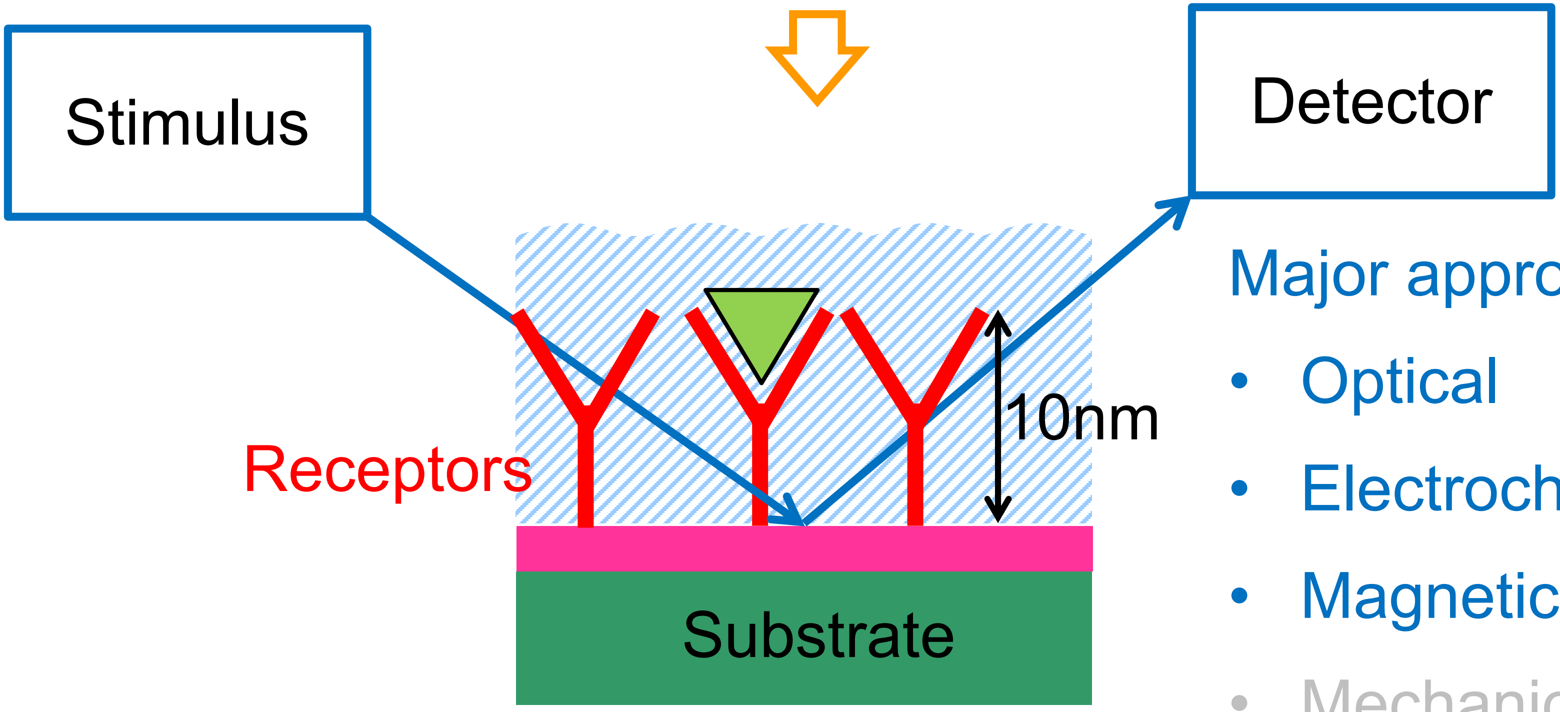
Detection Approaches

Extremely **thin** sensitive layers, in liquid environment

Detection tecniques sensitive to planar surface changes



Major approaches:

- Optical
- Electrochemical
- Magnetic
- Mechanical

Everything happens at the **interface!**

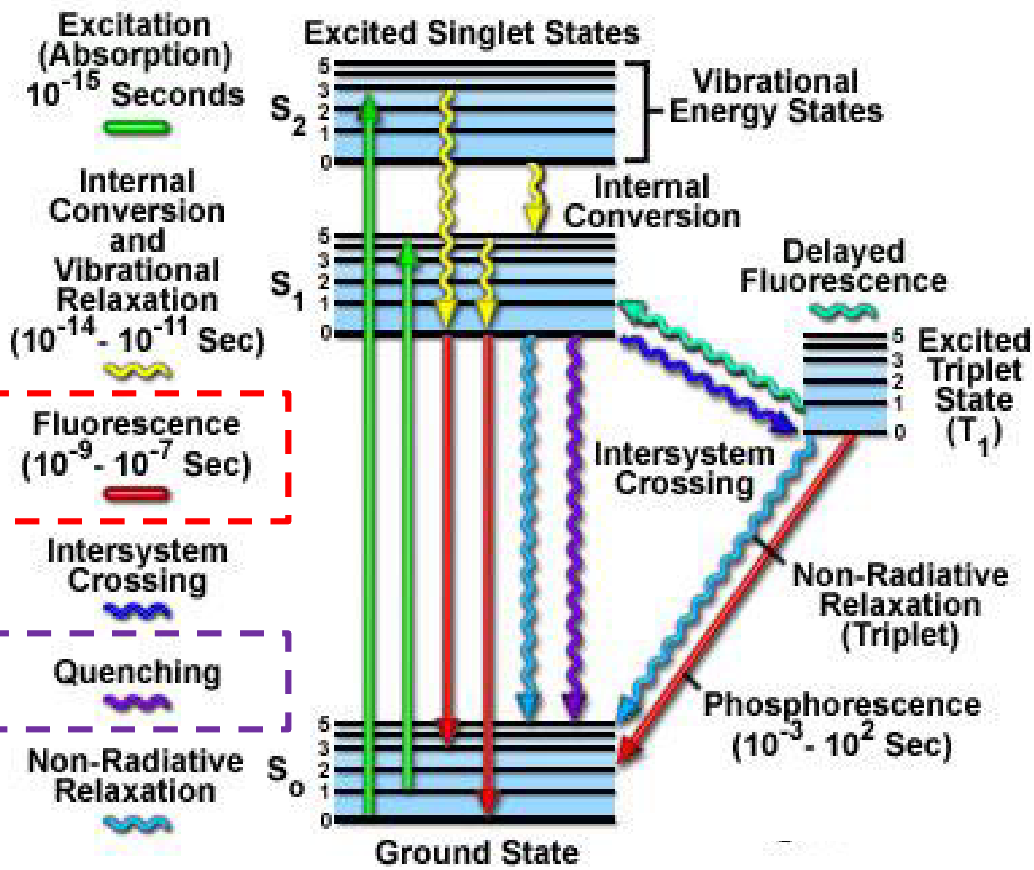
Optical Detection

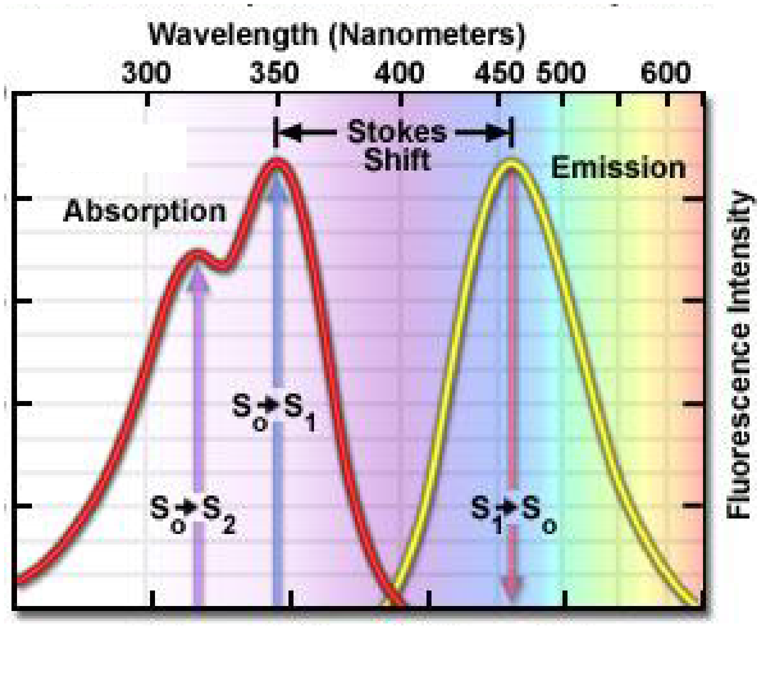
Optical microscopy is the preferred detection tool in Biology
non invasive, relatively high power, molecularly selective

1. Fluorescence: GFP, immunostaining
2. Scanning probe microscopy
3. Sources, filters, detectors
4. Solid-state integrated detection: Silicon Photonics

Fluorescence

Light emission by optically excited molecules.

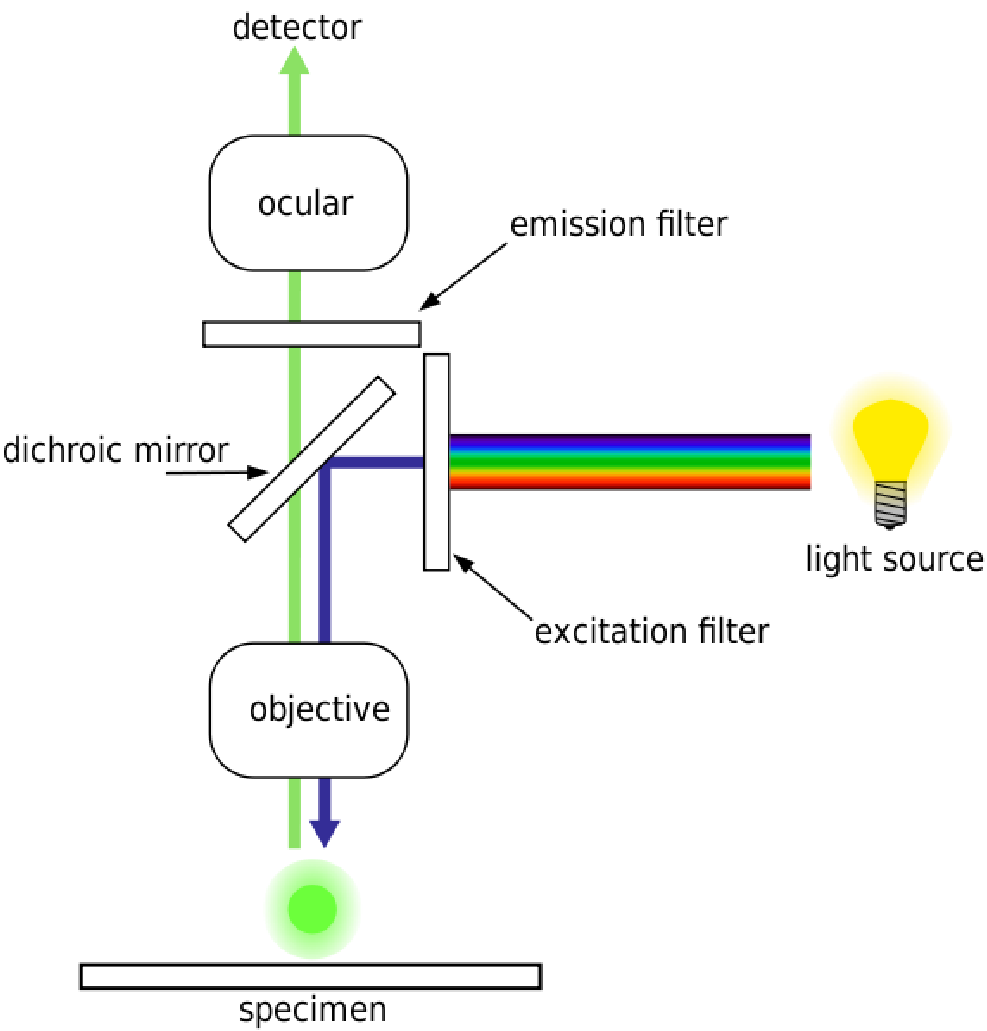




A large energetic separation (*Stokes shift*) makes easier the detection

Bleaching limits the number of excitation cycles

Fluorescence Microscope



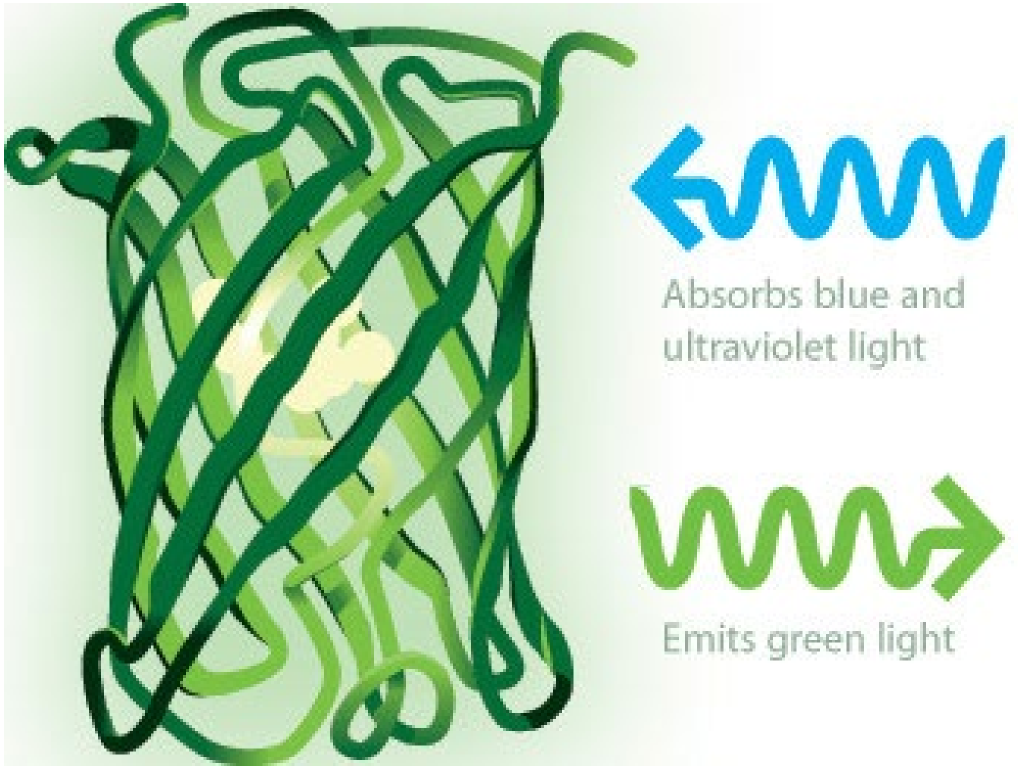
Optical microscopy is the workhorse of laboratory-scale Biology research.

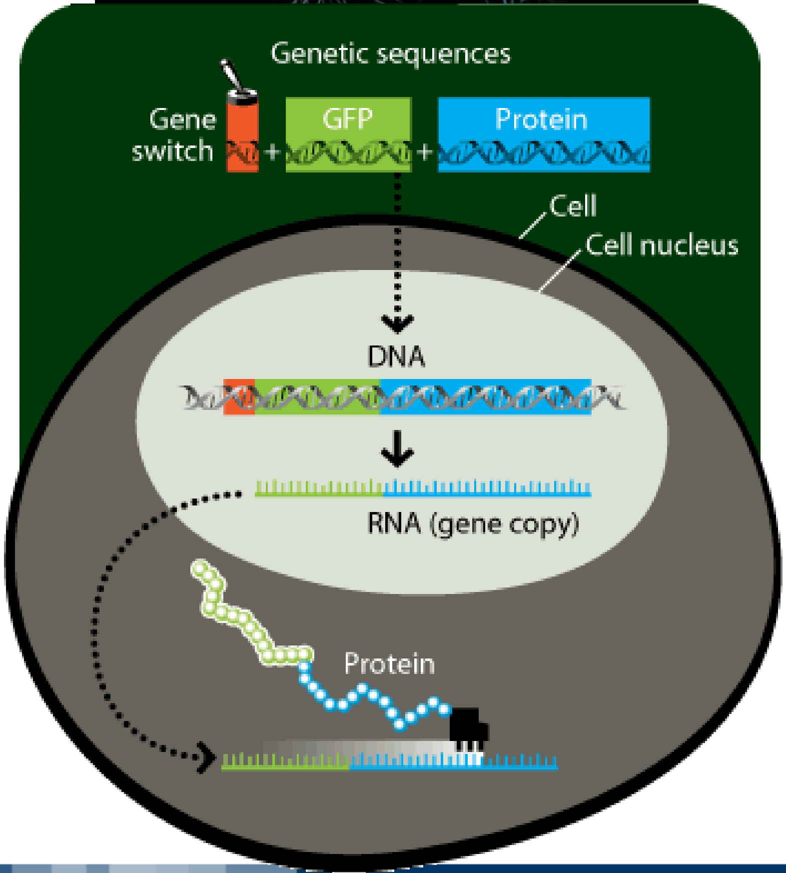
Since cells are transparent, fluorescence microscopy is used to detect sub-cellular entities.

It can report the presence of a given molecule bond to the fluorescent molecule.

Optical Marker: Green Fluorescent Protein

Fluorescent protein of natural origin
used as a photo-activable
label of genetic expression

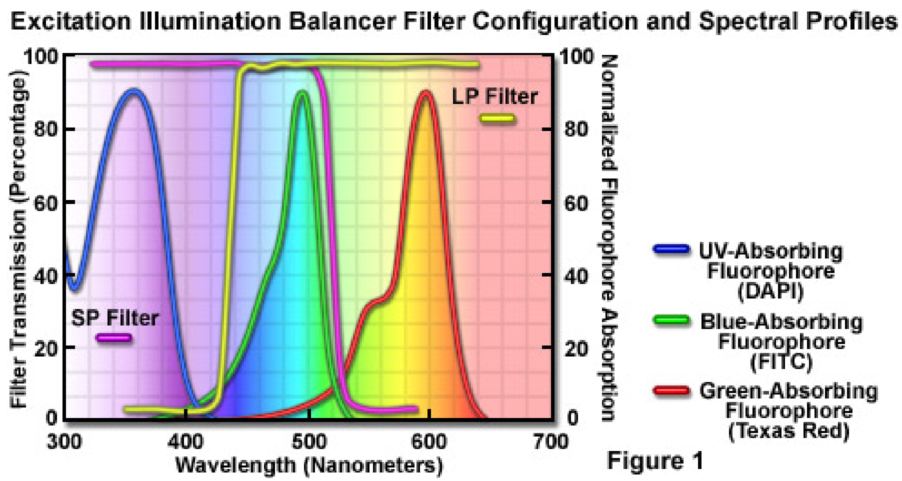


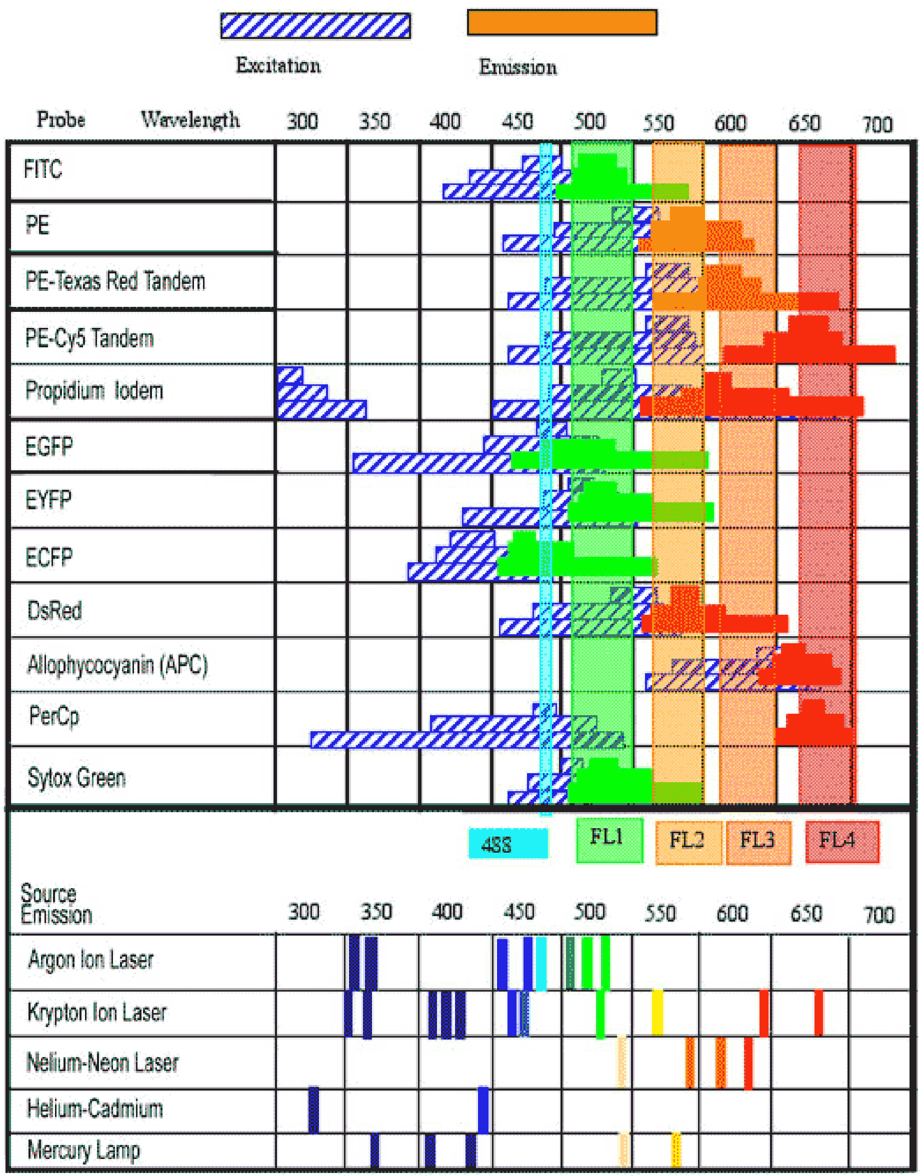


Different Molecules and Filters

Engineered more molecules to provide more colors.

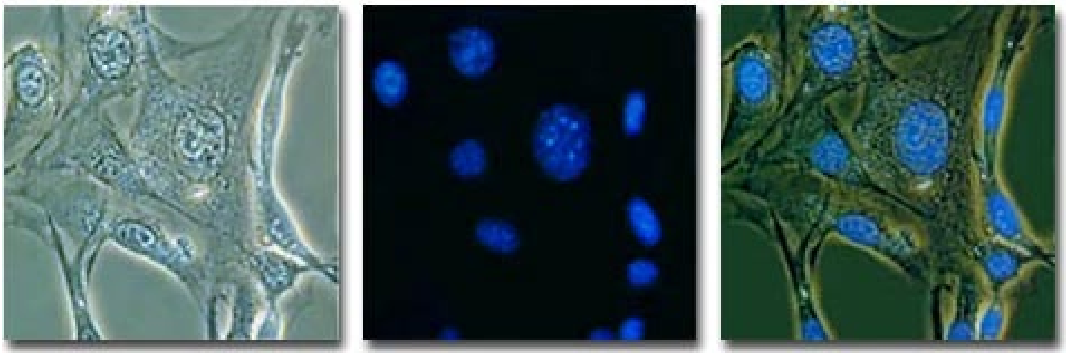
The microscope has a set of corresponding filters (R,G,B)

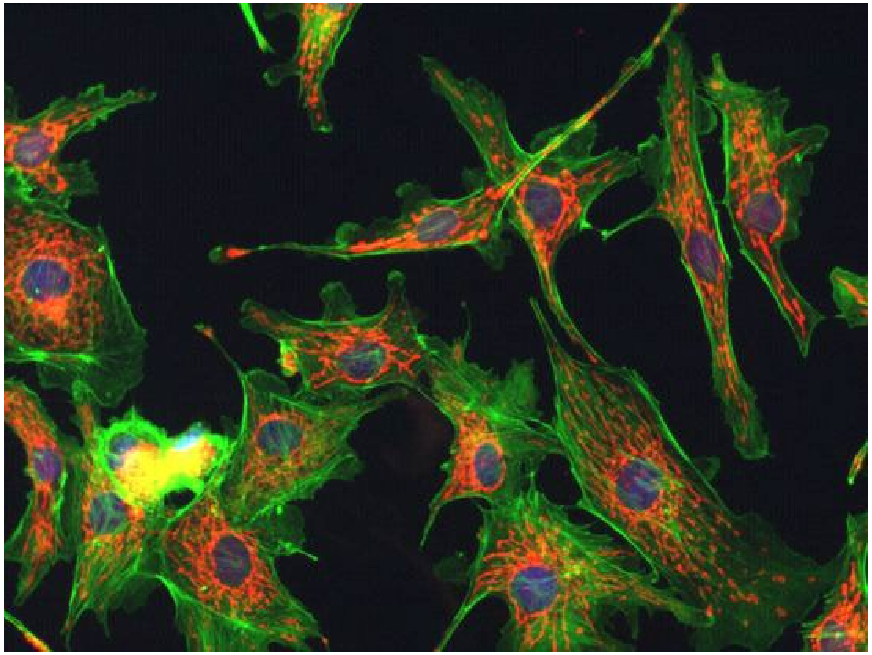




Examples

Bright Field Fluorescence Combined



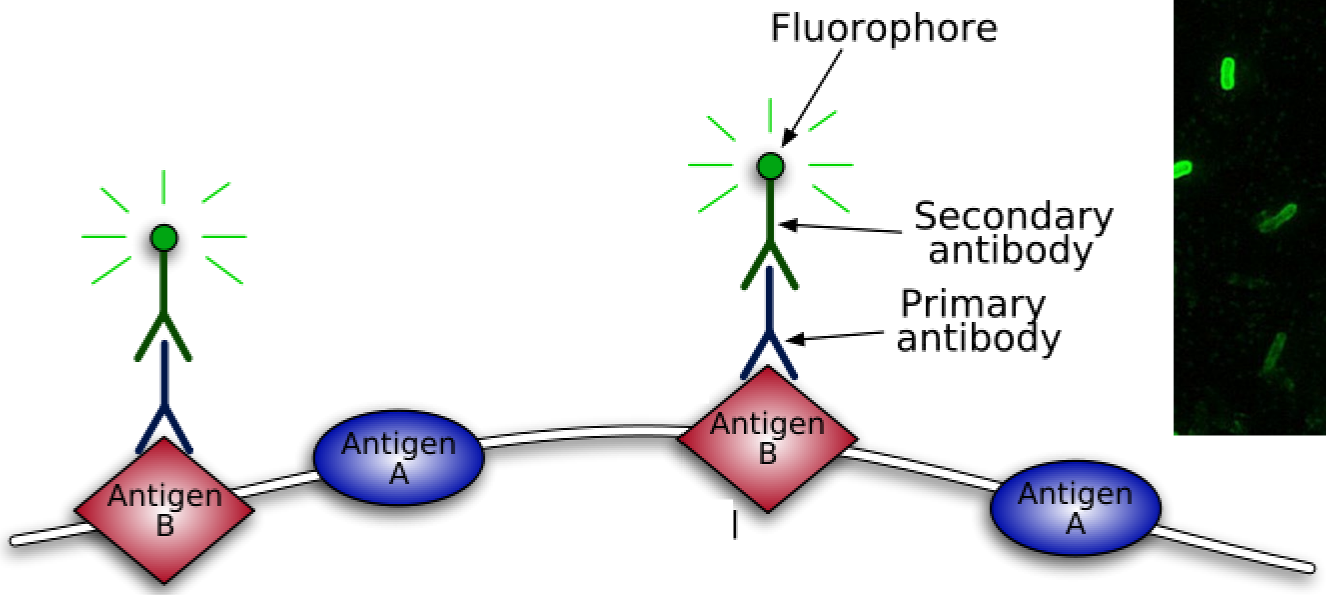


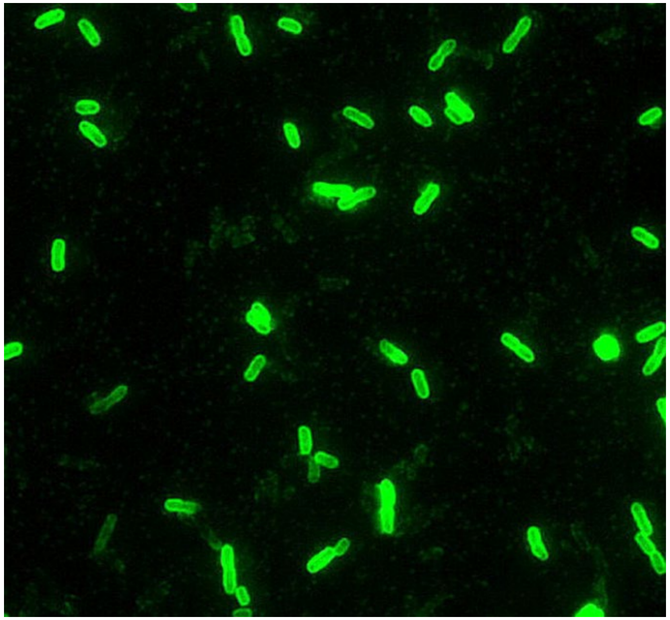
Selective staining: penetrate the cell membrane and bind to specific parts:

- Blue: nucleus
- Green: cytoskeleton
- Red: mitochondria

Molecular Markers Detection: Immunostaining

The optical label is attached to an antibody that **specifically** binds with a target molecule (antigen), whose presence can be optically detected.

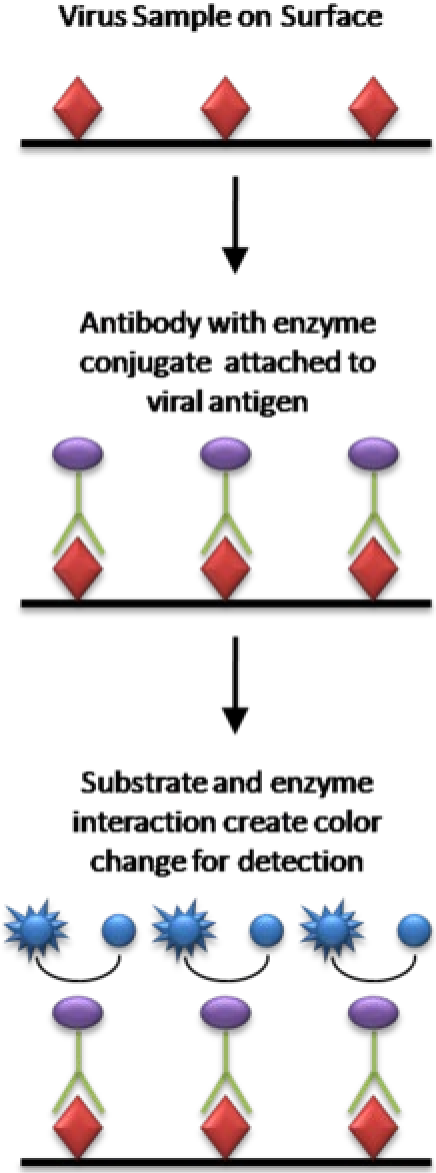




ELISA

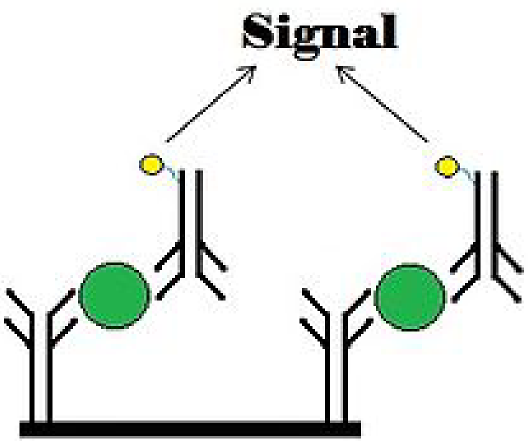
Enzyme-Linked Immunosorbent Assay

Standard biochemical test: the color change is proportional to the quantity of the enzyme



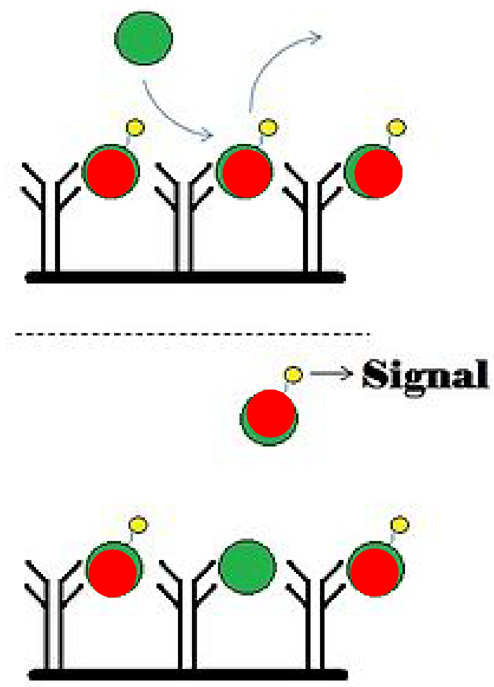
Types of Assays

Sandwich



Optical (fluorescent or visual) signal is proportional to the quantity of bound analytes

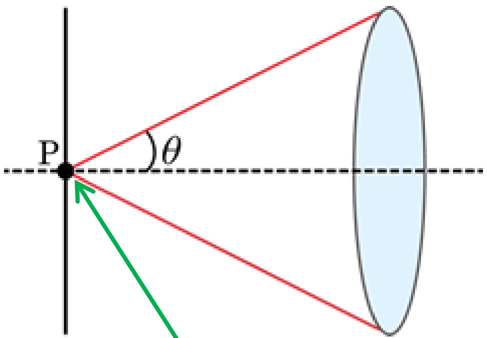
Competitive

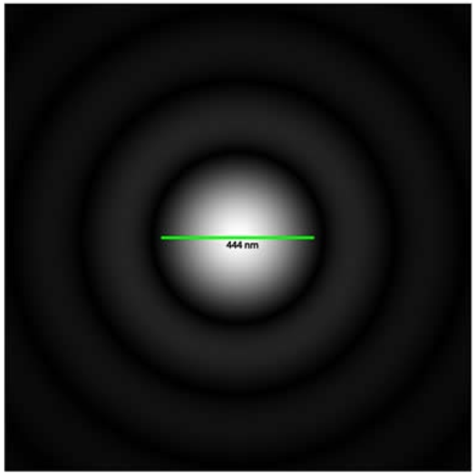


Signal decreases proportionally to the analytes replacing the bound markers

The Limits of the Optical Microscope

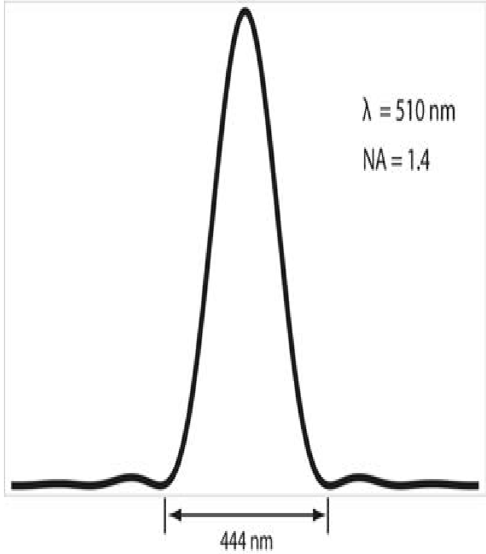
The spatial resolution is limited by diffraction:

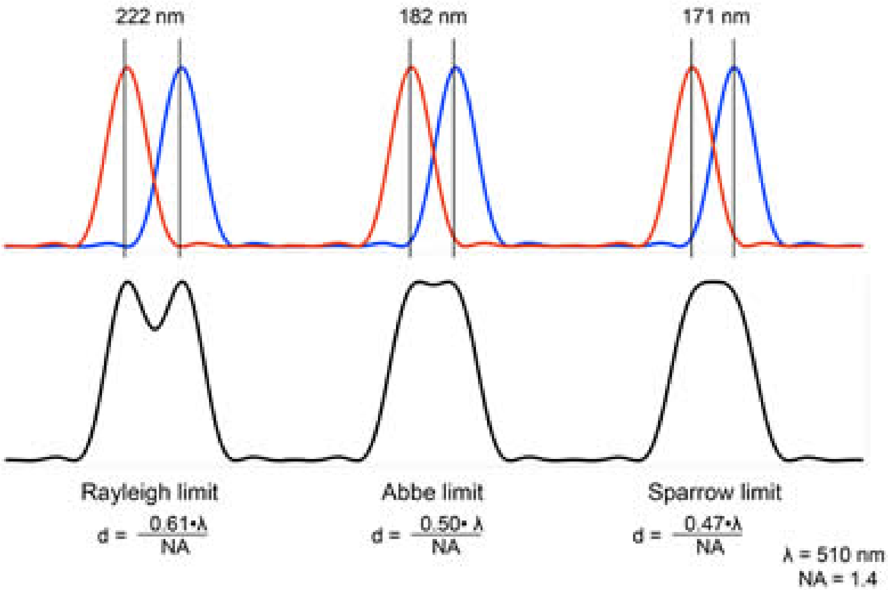




$$r = \frac{0.5\,\lambda}{NA} = \frac{0.5\,\lambda}{n\,sin(\theta)}$$

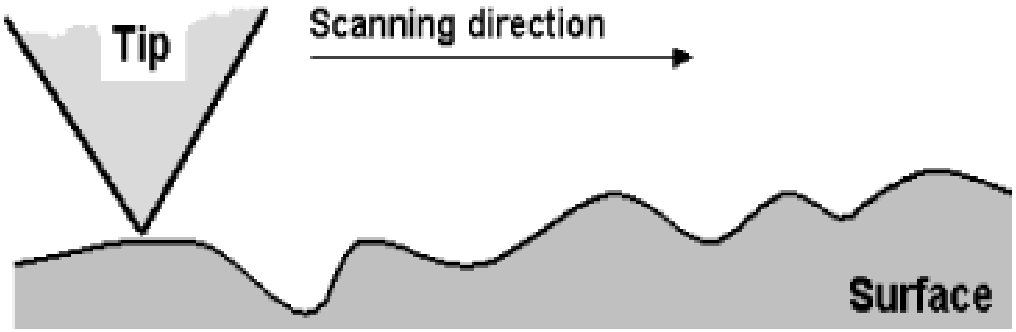
Single fluorophore (4 nm) is a point source

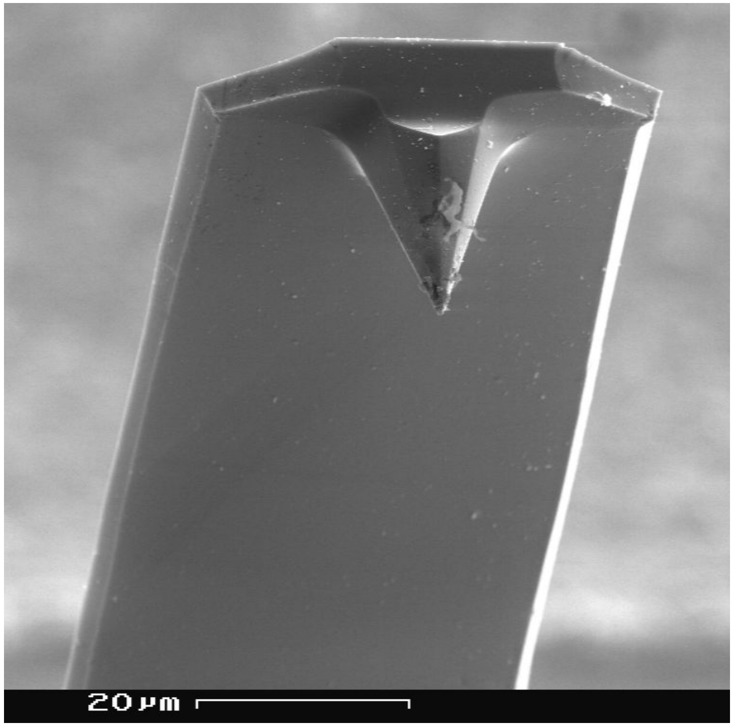


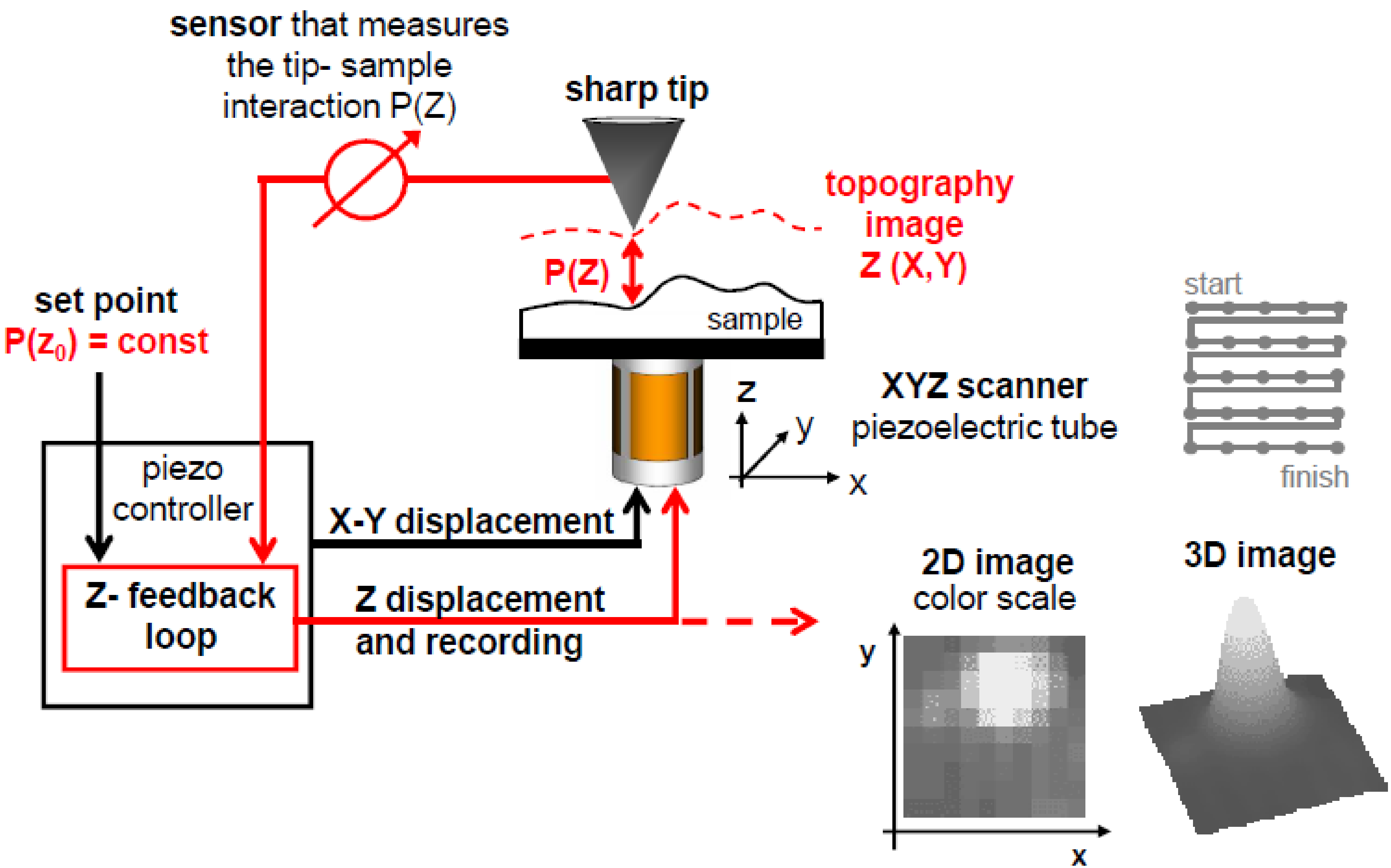


Beyond The Limits of Diffraction

Surfaces can be "imaged" below the optical resolution limit, by scanning them with a sharp microfabricated tip, interacting with the surface atoms via short-range **forces** whose intensity is spatially mapped







sensor that measures
the tip- sample
interaction P(Z)
sharp tip
topography
image
Z (X,Y)
P(Z)
sample
set point
$P(z_0)$ = const
start
finish
Z
y
x
XYZ scanner
piezoelectric tube
piezo
controller
X-Y displacement
Z- feedback
loop
Z displacement
and recording
2D image
color scale
3D image
y
x

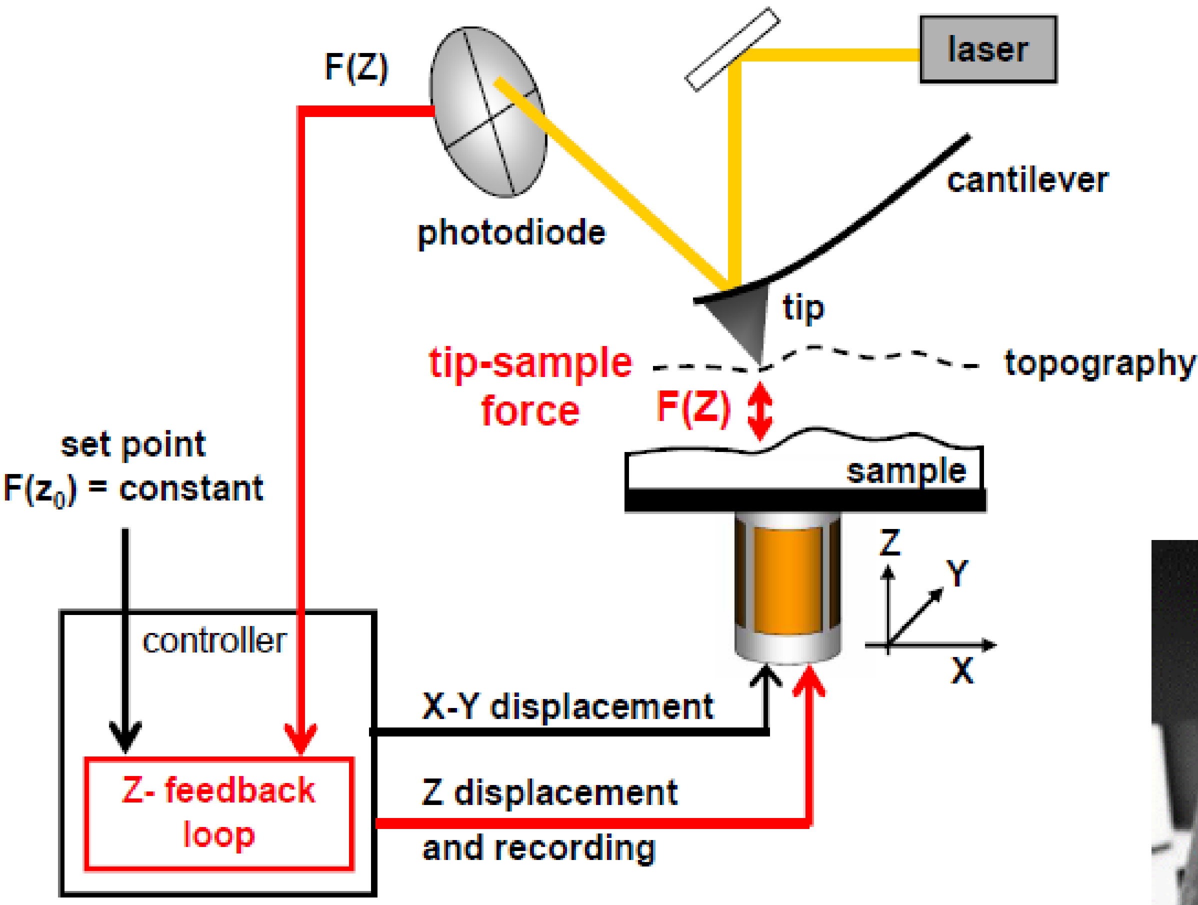
A Family of Techniques

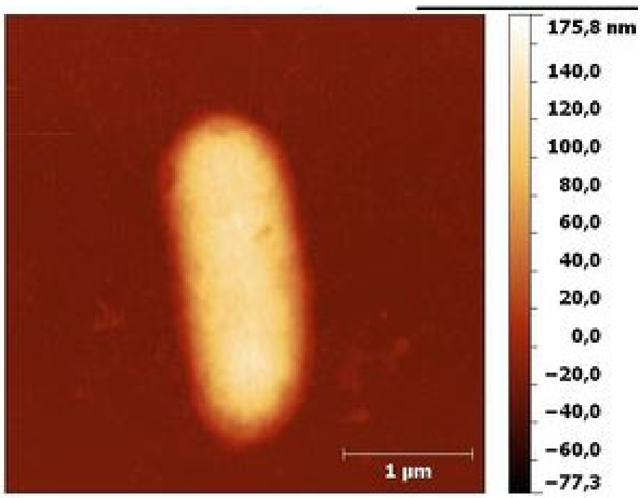
- **Scanning Tunneling Microscopy (STM) - 1982**
  - Scanning Near-Field Optic Microscopy (SNOM) - 1984
  - Atomic Force Microscopy (AFM) – 1986
    - Magnetic Force Microscopy (MFM)
    - Electrostatic Force Microscopy (EFM)
      - Kelvin Probe Force Microscopy (KPFM)
    - Current-Sensing AFM (CS-AFM)
      - DC: Conductive-AFM (C-AFM)
      - AC: Nanoscale Impedance Microscopy (NIM)
  - Scanning Capacitance Microscopy (SCM) – 1985
  - Scanning Electrochemical Microscopy (SECM) - 1989
  - Scanning Ion Conductance Microscopy (SICM) - 1989

Binnig and Rohrer,

Nobel Prize in Physics, 1986

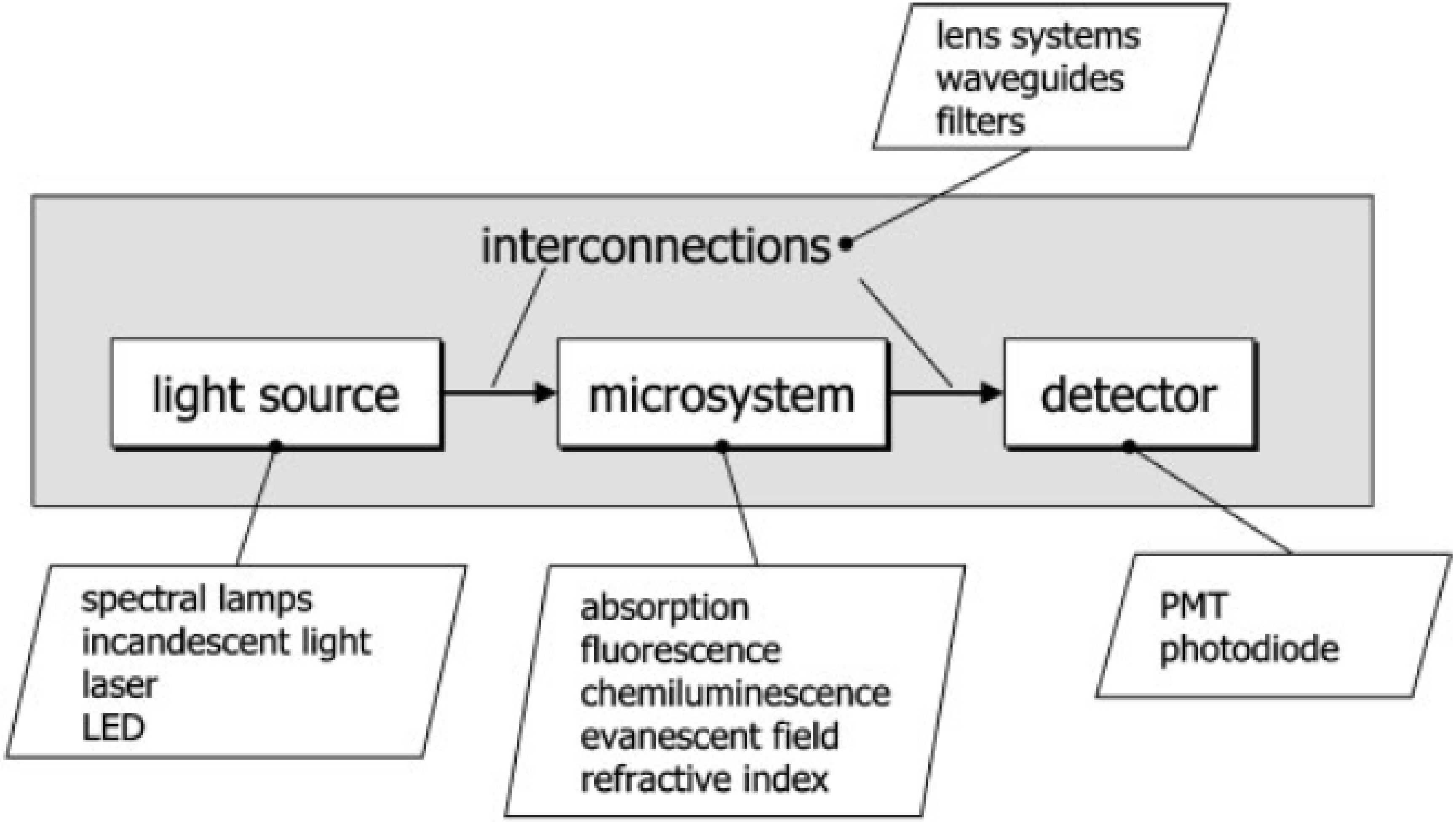
The Atomic Force Microscope





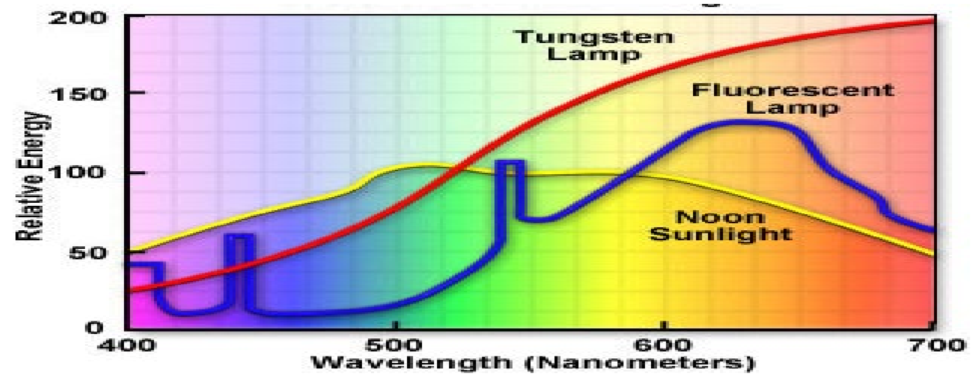
… like reading Braille!

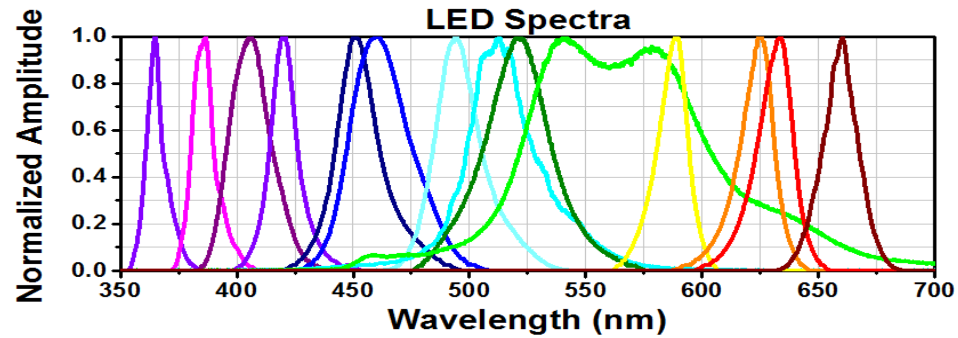
Optical Detection in Biochips

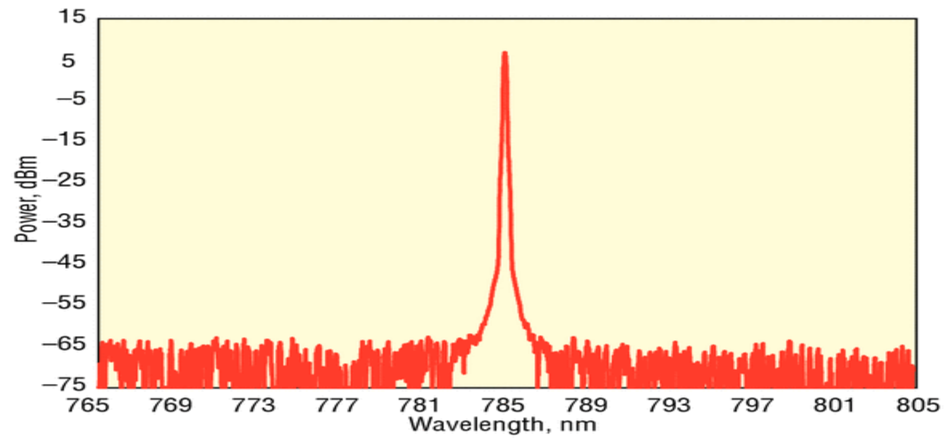


Light Sources

- **Lamps**
  - High power
  - High dissipation
  - Wide spectrum (filter)
- **LEDs**
  - Solid-state
  - High efficiency
  - Limited power (growing)
- **Lasers**
  - Monochromatic
  - Solid-state
  - cost/size, few λ



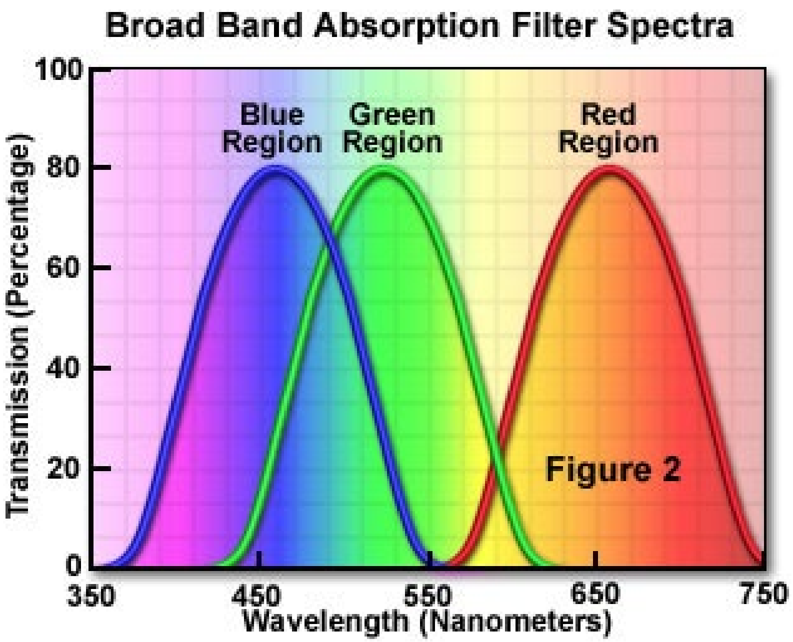


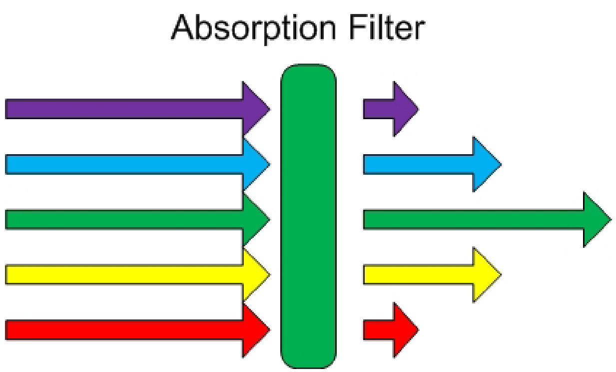


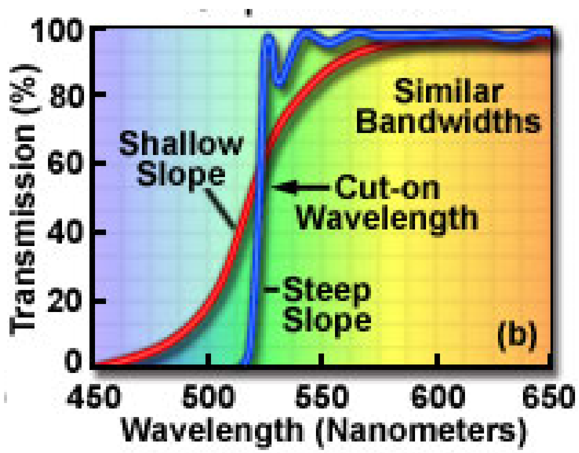
Absorption Optical Filters

Colored (pigmented) films that absorb (i.e. convert into heat) some wavelengths

- Large bandwidth
- Shallow slopes







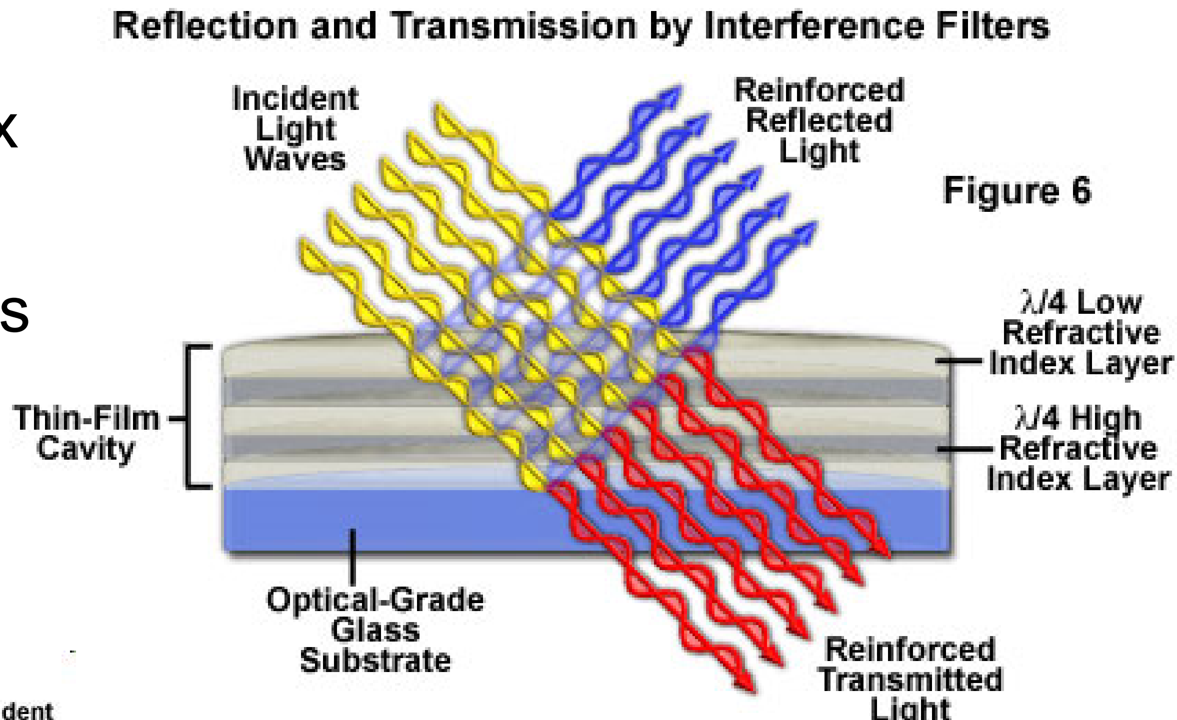
Interference Filters

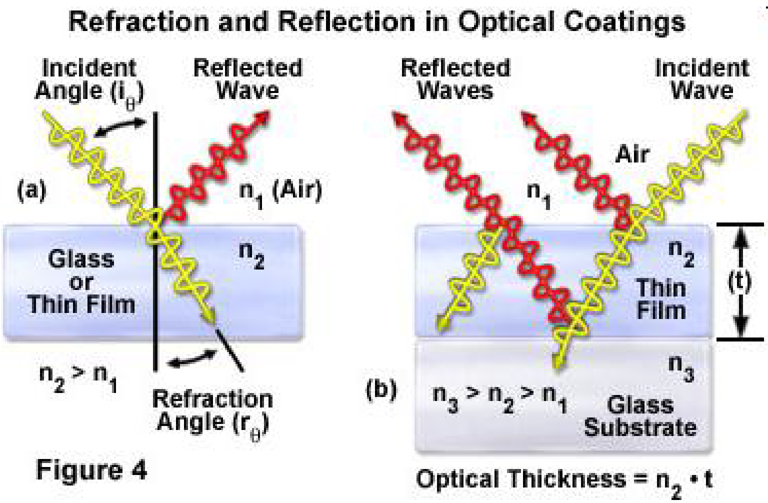
Stack of thin layers with alternate refraction index

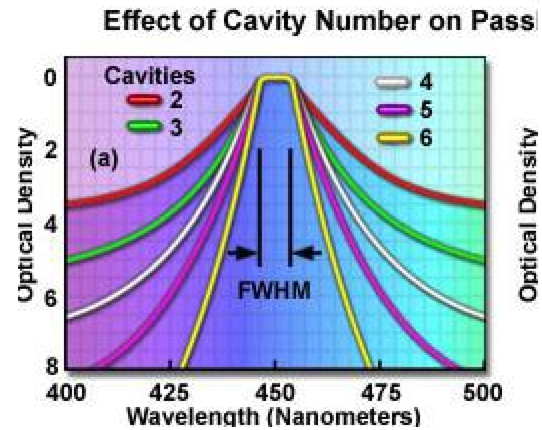
- Controllable properties

 (nm deposition)

- Sharp slopes

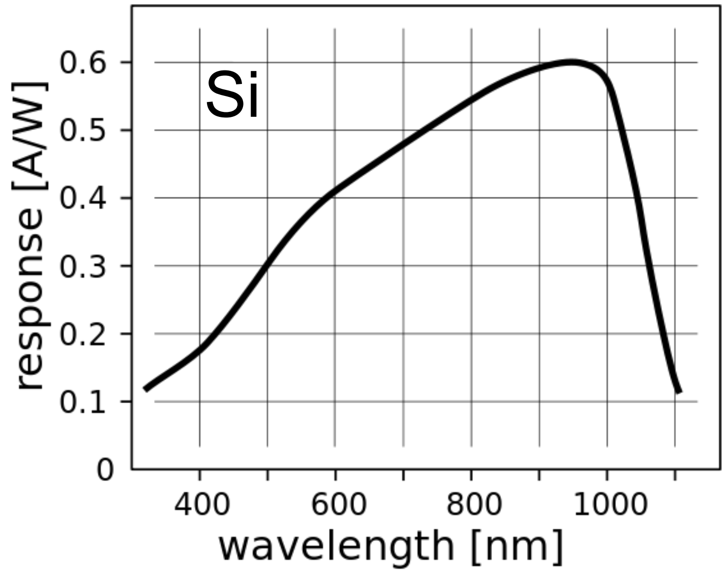


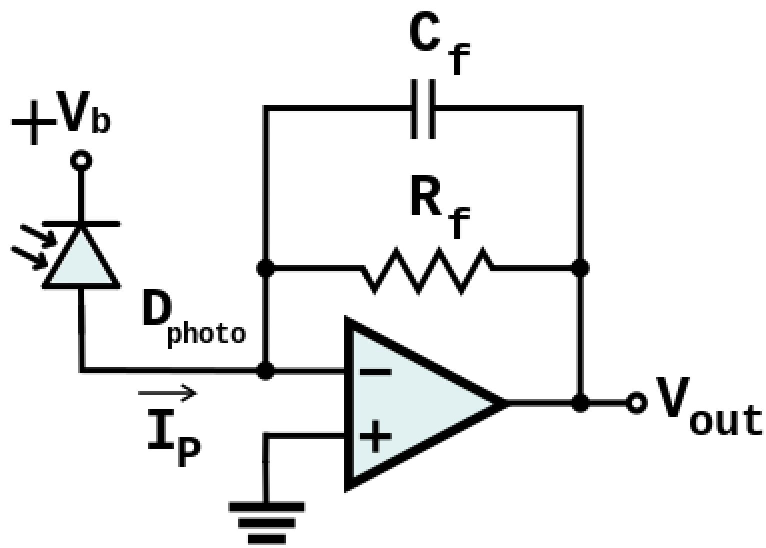




Light Detectors

- Photoresistors
- **Photodiodes:**
  - PN
  - PIN
- Avalanche photodiodes:
  - Photomultipliers (PMT)
  - APDs
  - SPADs
  - SiPMs
- Arrays (image sensors):
  - CCD
  - CMOS





Detection Approaches for Biosensing

With external stimulation:

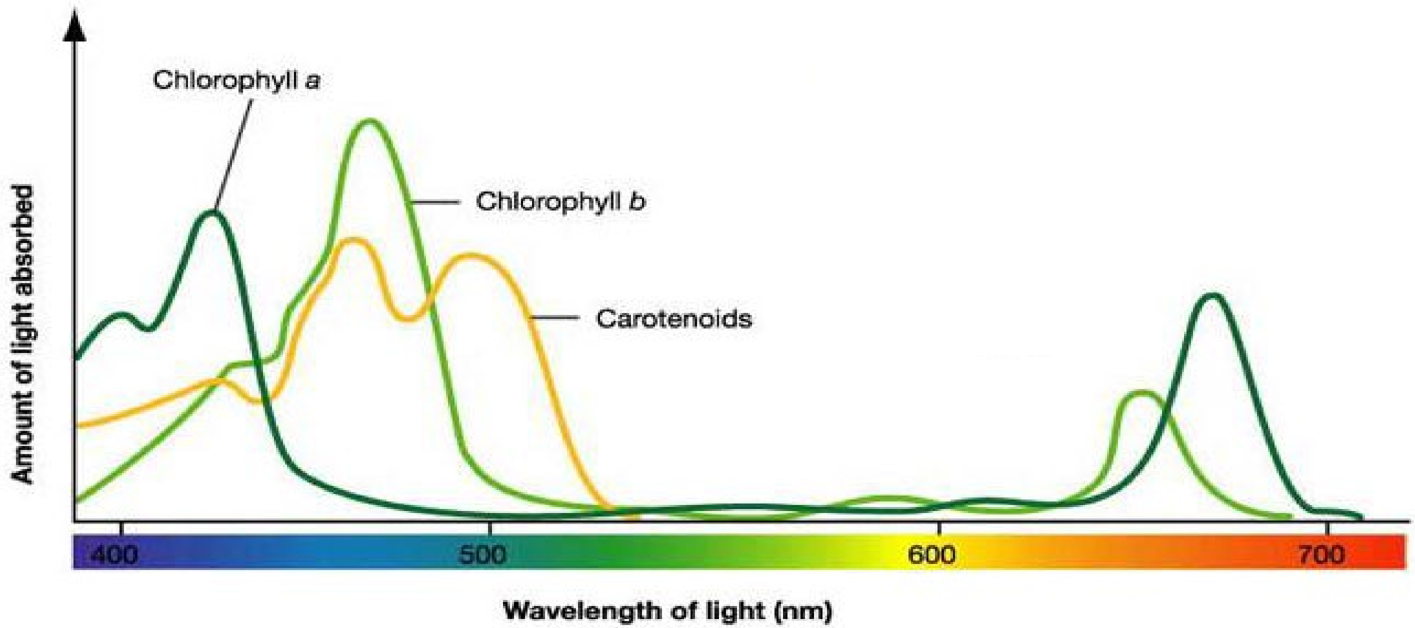
- Light absorption (light extinction)
- Fluorescence (energy transfer)
- Evanescent field interactions (light propagation)
- *Surface Plasmon Resonance*: sophisticated technique very sensitive to the metal surface properties.

No external source excitation:

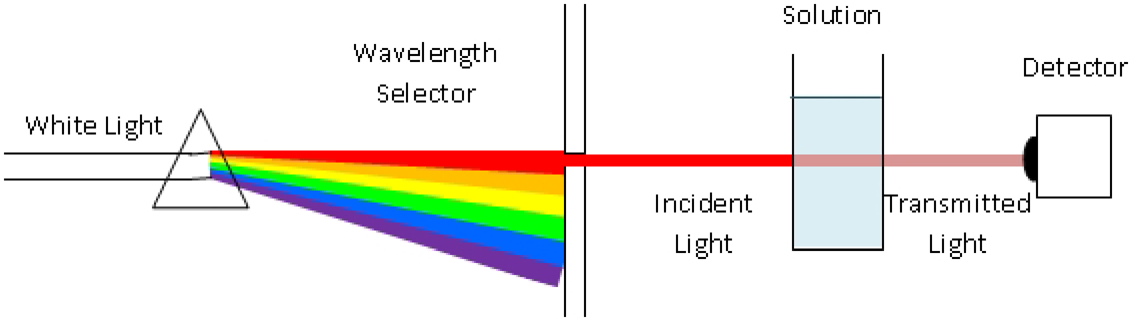
- *Chemiluminescence*: some biochemical reactions produce light as a product

Spectroscopy

Each compound has an absorbance spectrum (fingerprint):



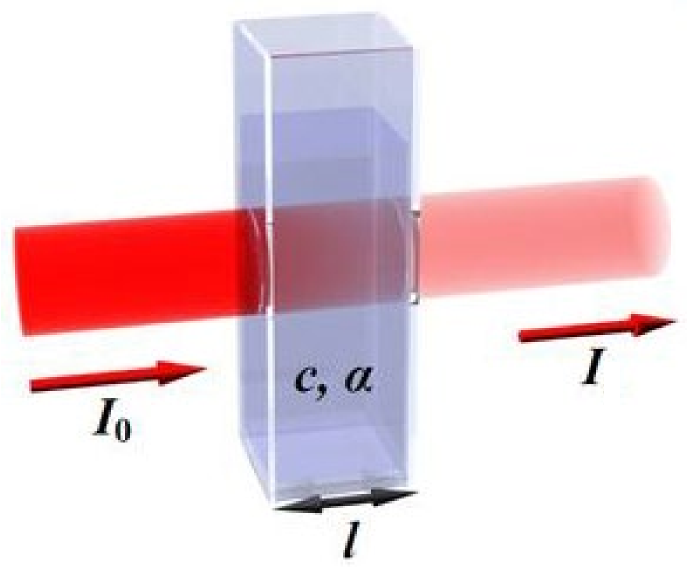
Spectrophotometer:



Requires a dispersive medium (prism, grating) and a broadband detector (or an array)

Absorbance

At a single wavelength:



Lambert-Beer Law

Absorption:

$$A = \log_{10}\left(\frac{I_o}{I}\right) = \log_{10}\left(\%T^{-1}\right) = \varepsilon\, l\, c$$

α = A/l = absorbance

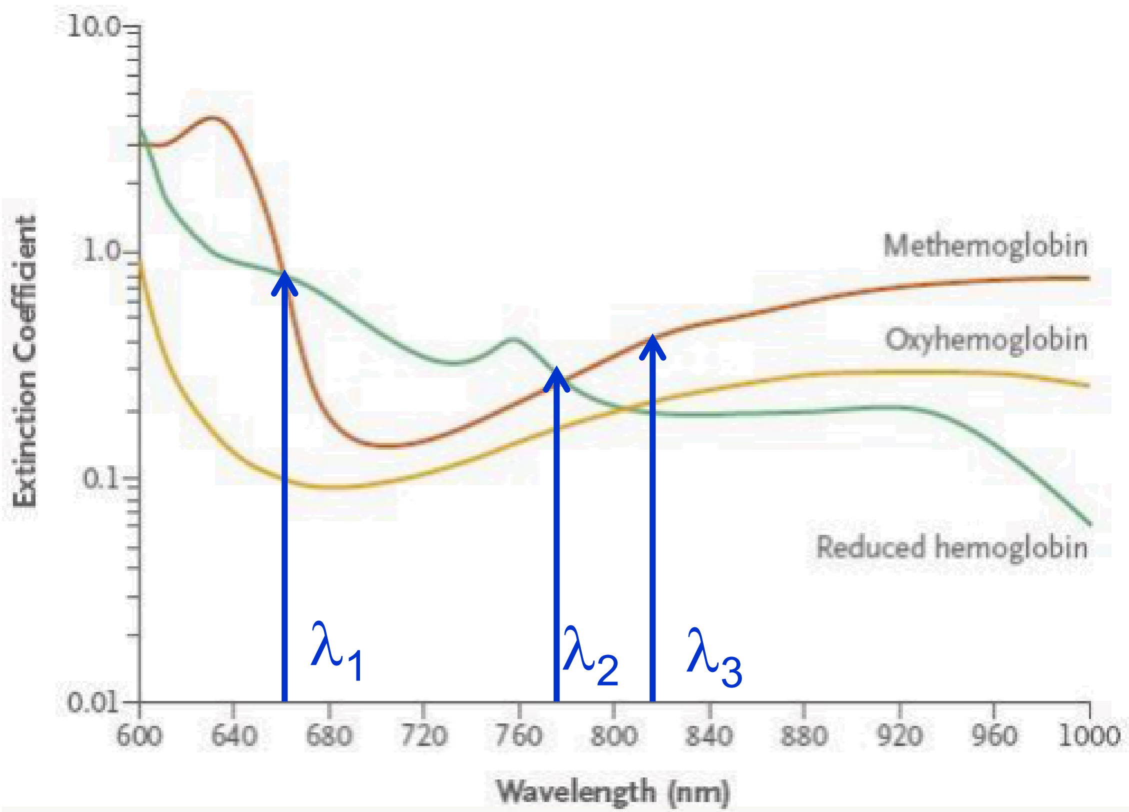
l = length

c = concentration of molecules

ε = molar absorbance

Example: Pulse Oximetry

$$\alpha(\lambda_1) = \varepsilon_{ox}(\lambda_1)[M_{ox}] + \varepsilon_{deox}(\lambda_1)[M_{deox}] + \varepsilon_{meth}(\lambda_1)[M_{meth}]$$
$$\alpha(\lambda_2) = \varepsilon_{ox}(\lambda_2)[M_{ox}] + \varepsilon_{deox}(\lambda_2)[M_{deox}] + \varepsilon_{meth}(\lambda_2)[M_{meth}]$$
$$\alpha(\lambda_3) = \varepsilon_{ox}(\lambda_3)[M_{ox}] + \varepsilon_{deox}(\lambda_3)[M_{deox}] + \varepsilon_{meth}(\lambda_3)[M_{meth}]$$

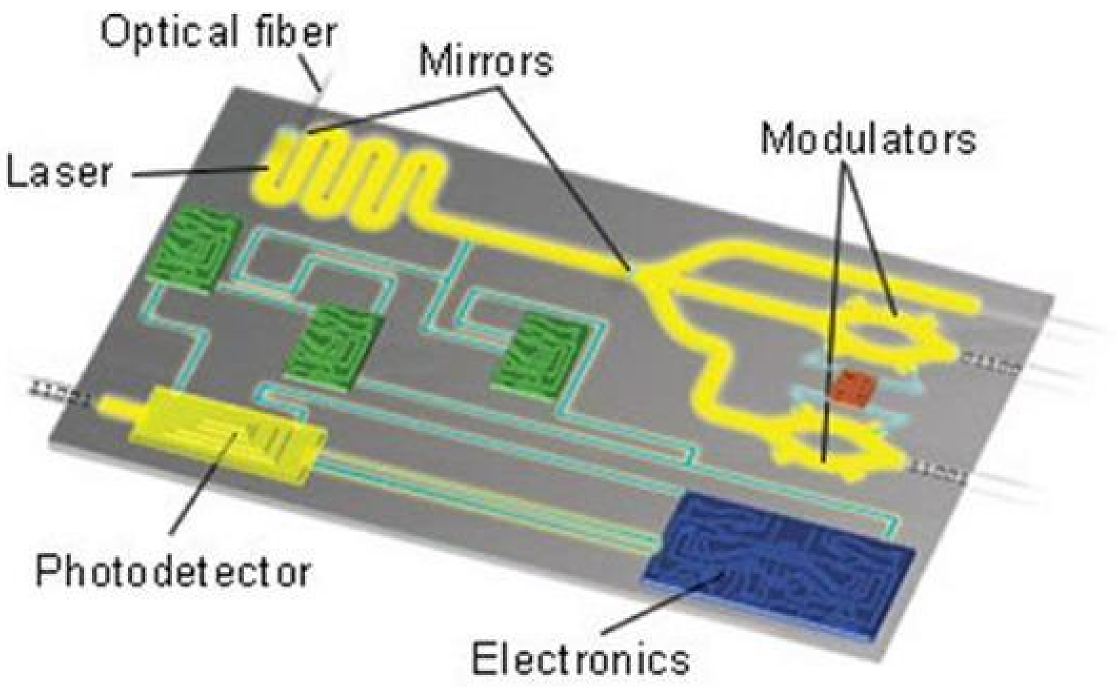


External monitor of O_2 saturation in blood

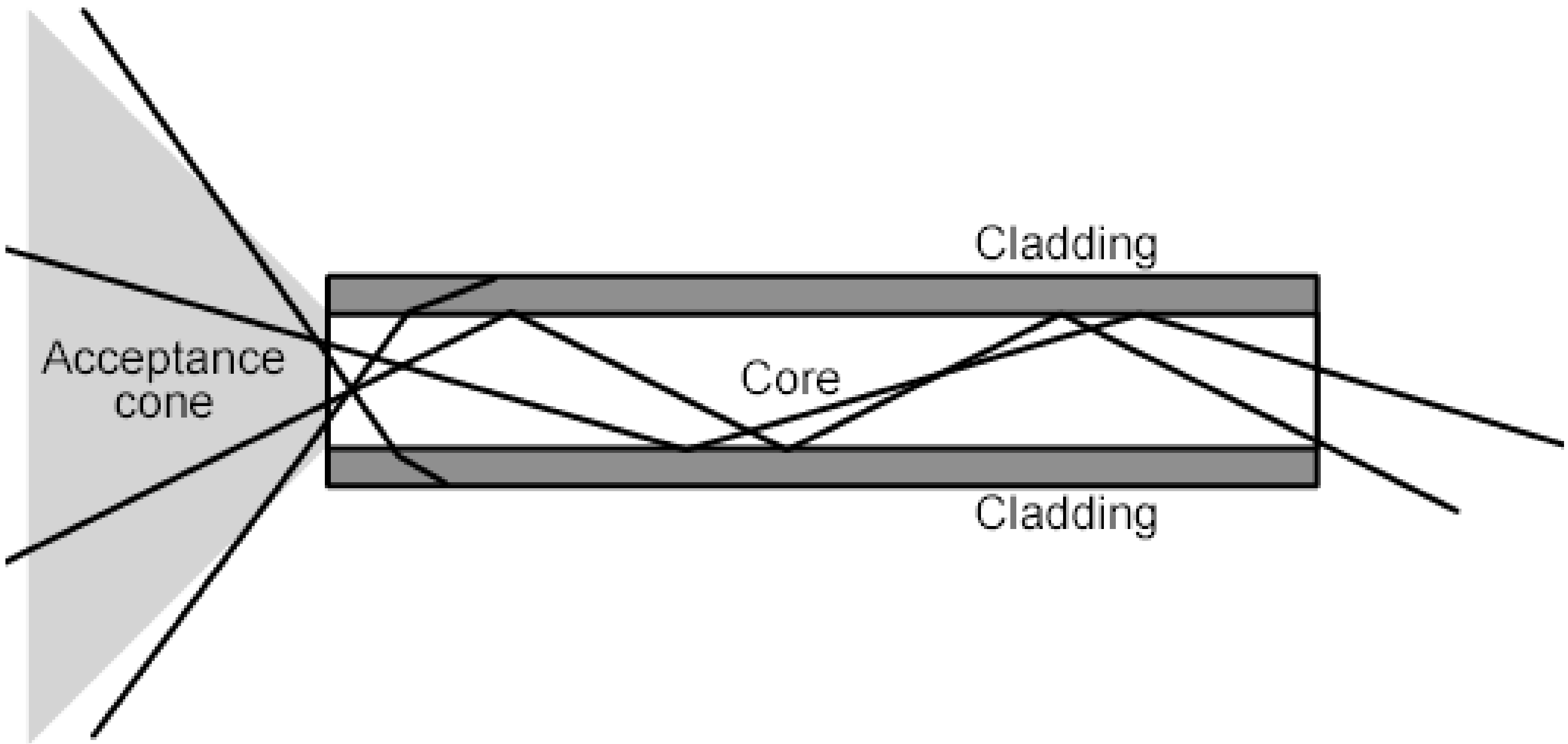
How to Handle Light on Chip: Silicon Photonics

Monolithic integration of sources, detectors, waveguides, switches, filters etc… on a single solid-state active (or passive) platform (Si, InP, Ge…)

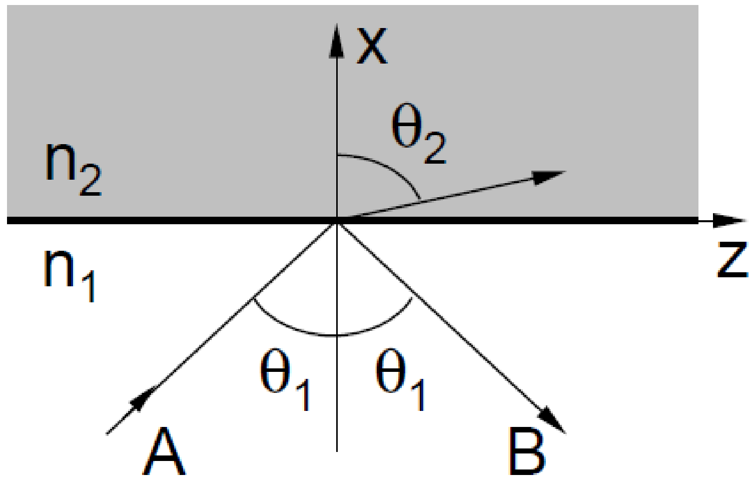
Applications: telecom (long and short distance), processing and also **biosensing**



Waveguides



Confined propagating EM field

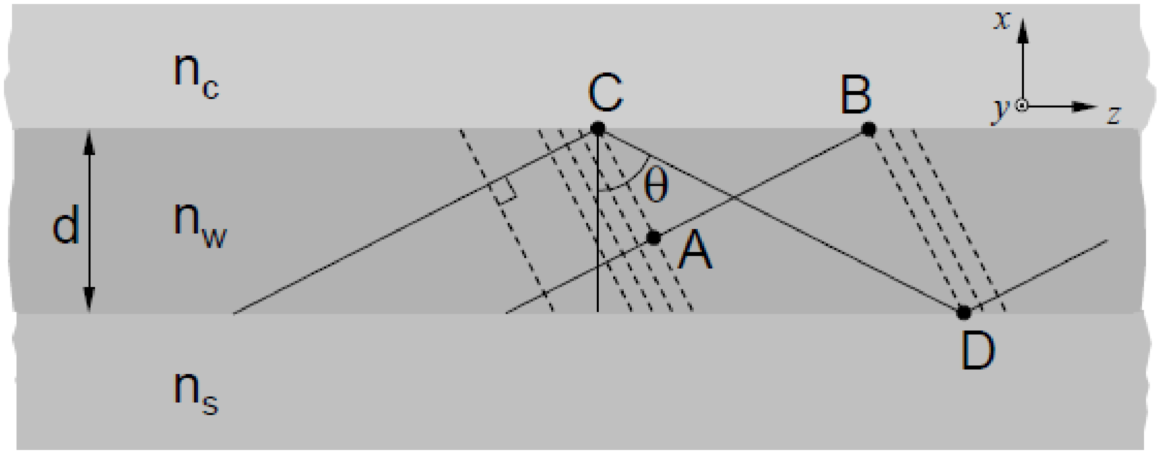


1. Total internal reflection condition

$$n_1 sin\theta_1 = n_2 sin\theta_2$$

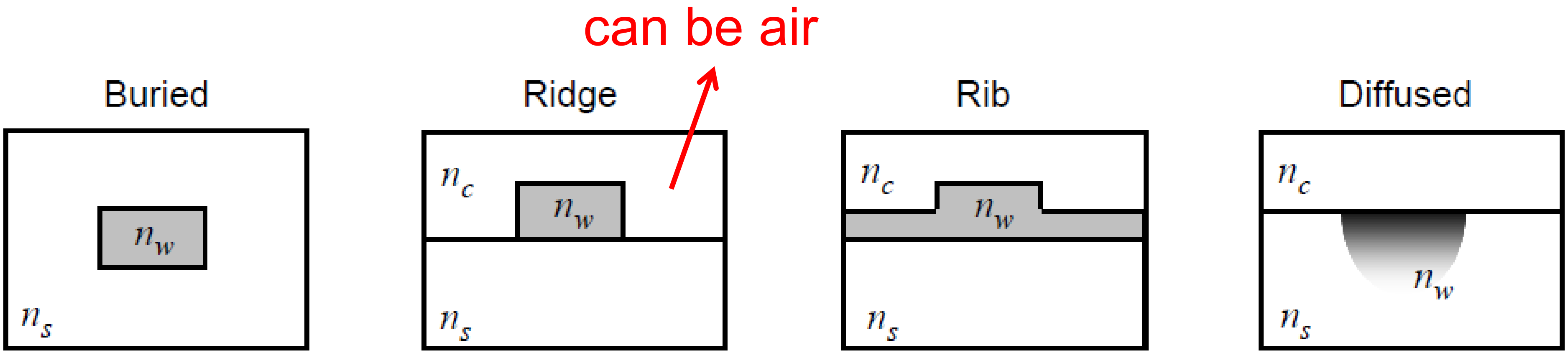
$$\theta_1 > asin\left(\frac{n_2}{n_1}\right)$$

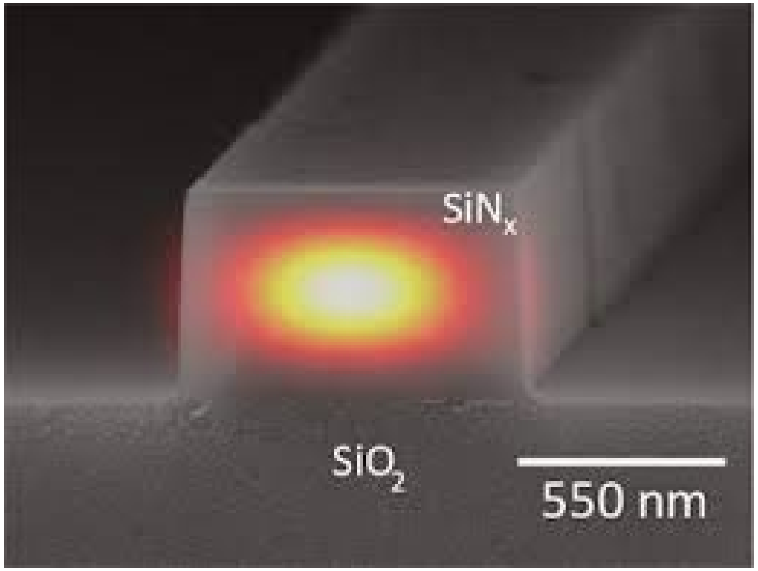
2. Phase coherence wavefronts



$$\frac{2\pi}{\lambda} n_w d cos\theta + 2\phi = N\pi$$

Integrated Waveguide Types

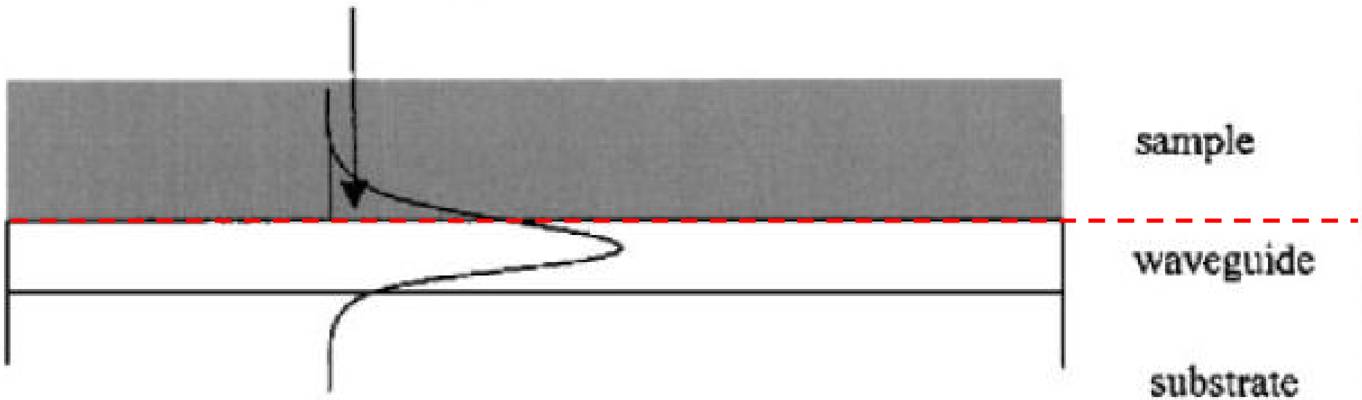


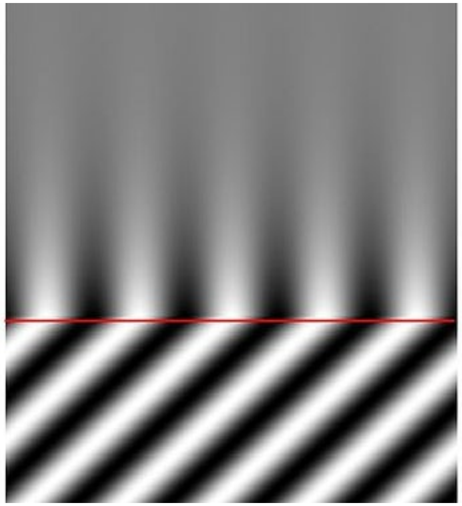


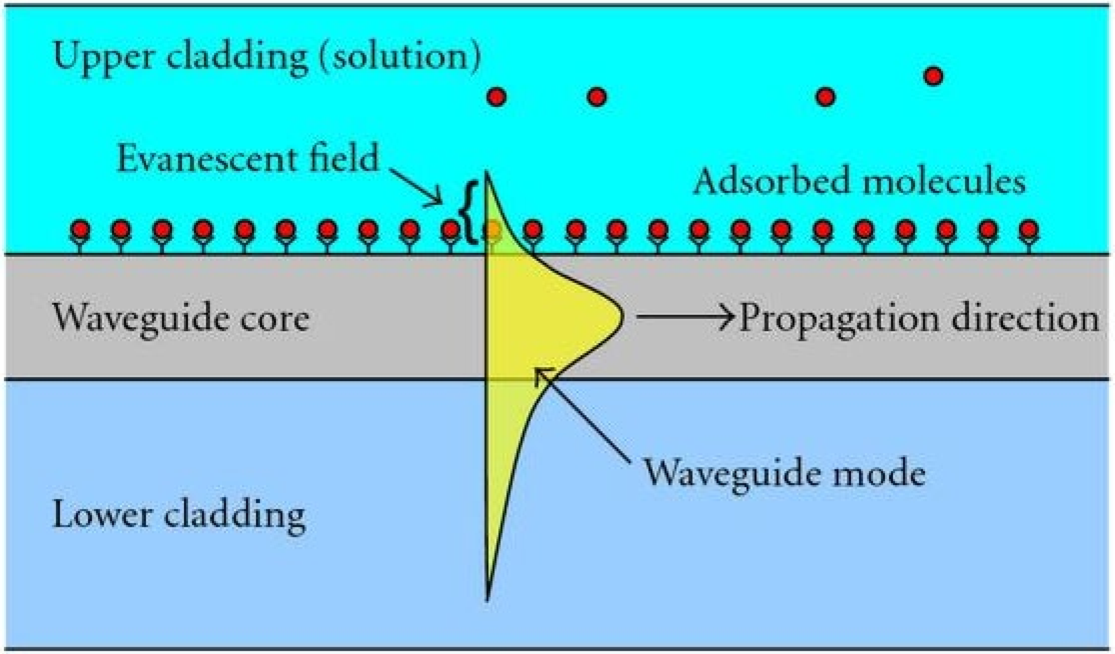
Materials:

- Core: Si, Ge, InP
- Cladding: SiO_2, air…

Evanescent Field





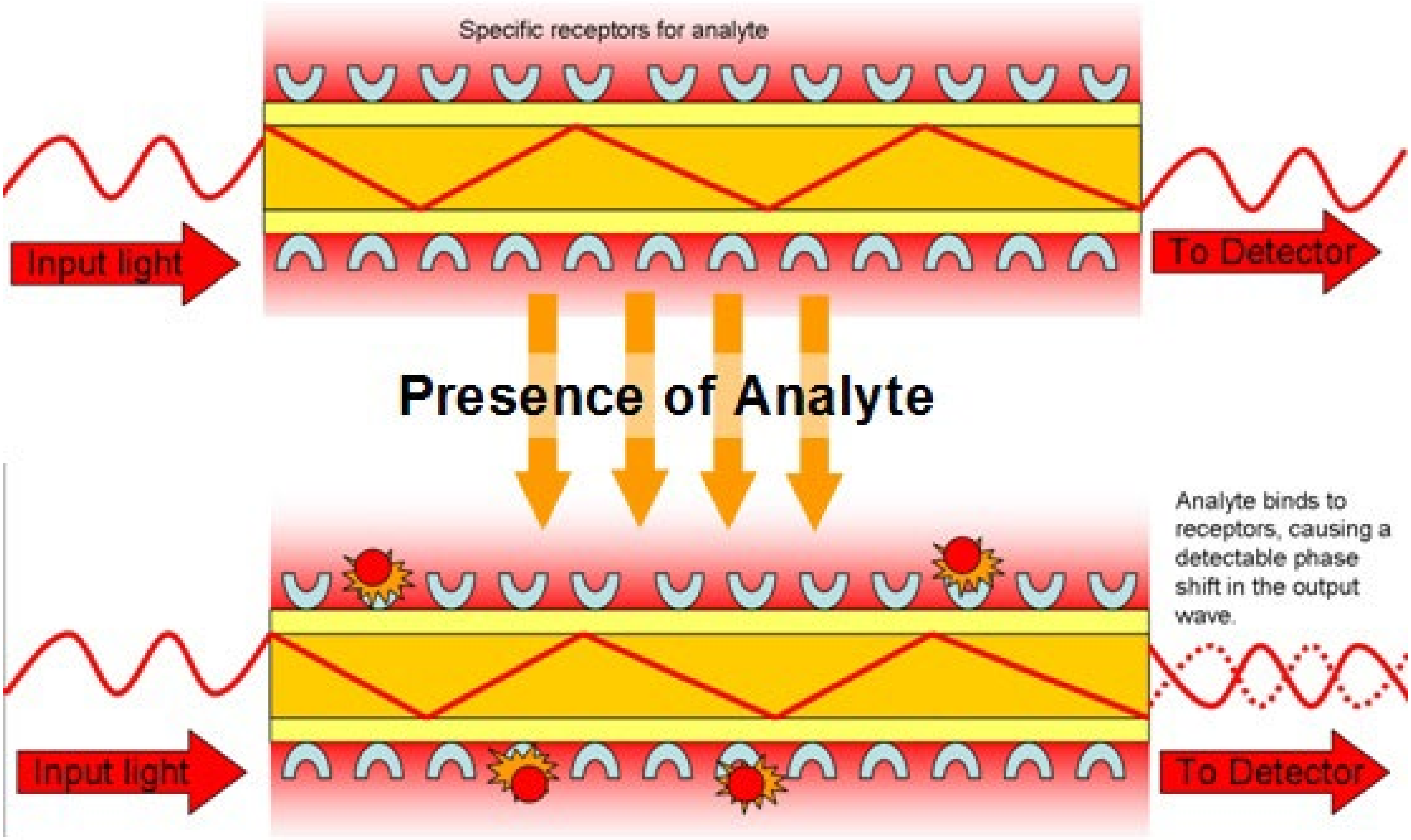


At total internal reflection,

Exponential decay of field intensity

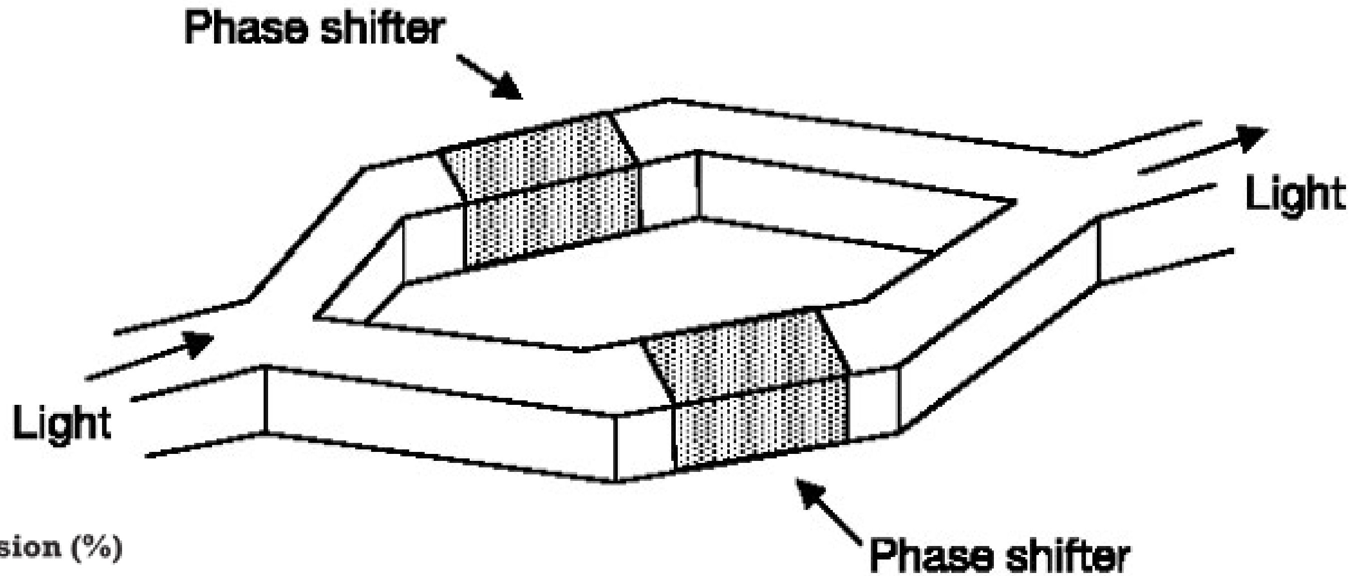
Propagating light is **affected by adsorbed molecules**

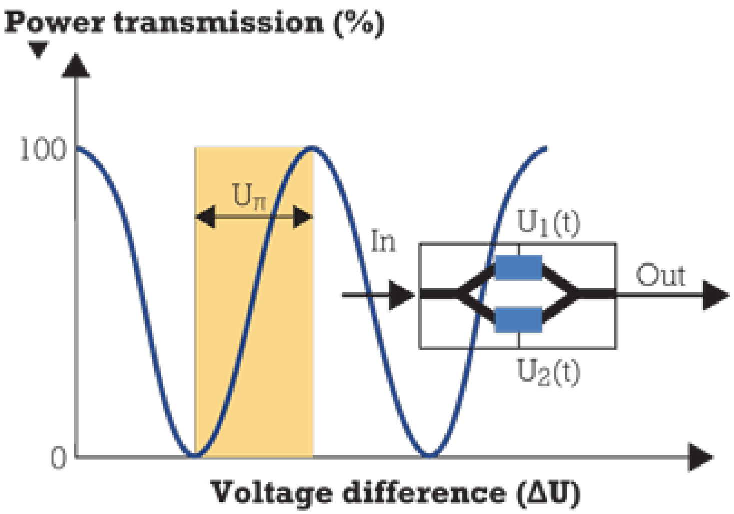
Waveguide as Biosensor



The Mach-Zehnder Interferometer

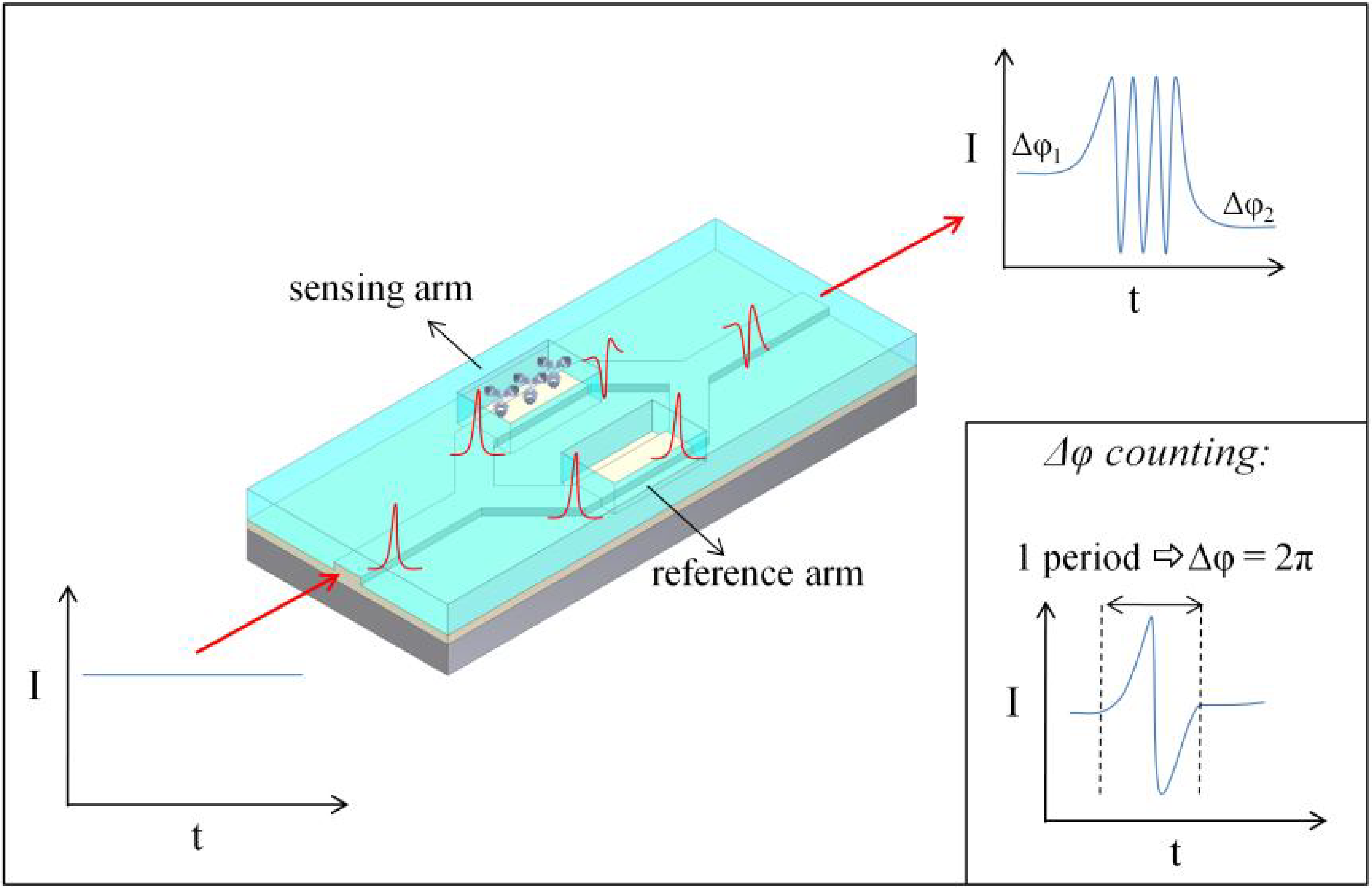
Useful to convert phase difference into intensity difference:



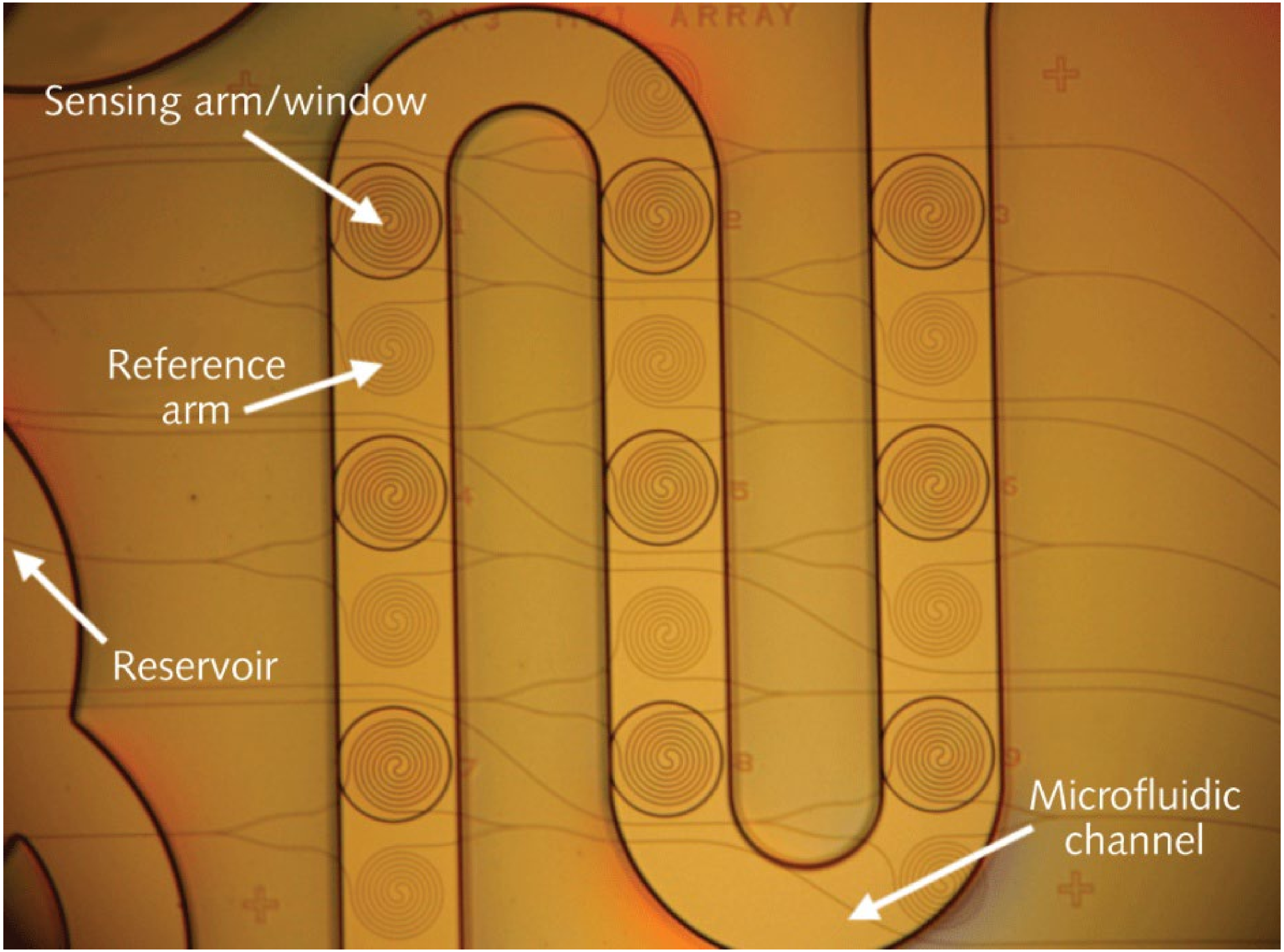


Light splits and combine after two different paths, whose optical length can be controlled (modulated)

Interferometer Branch Used as Biosensor

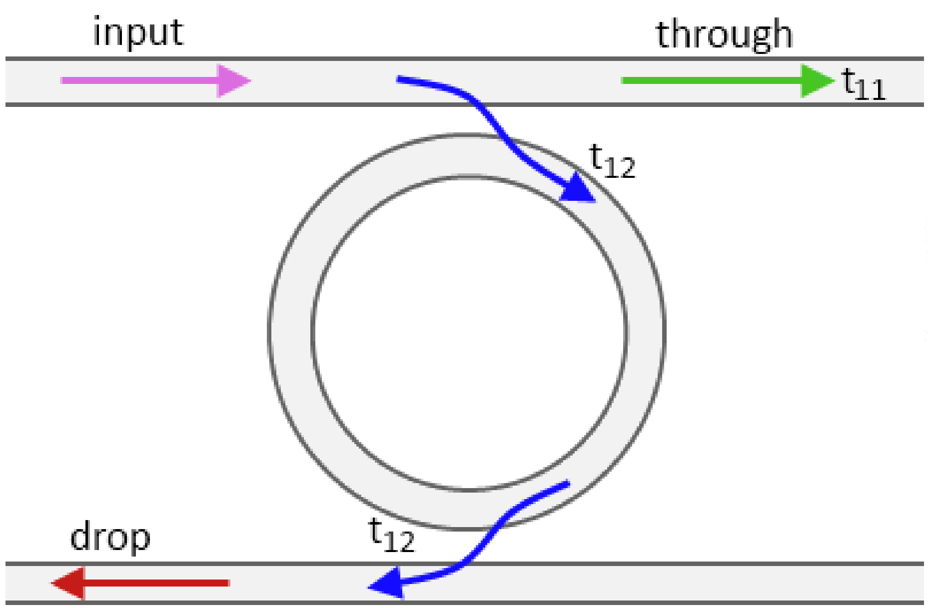


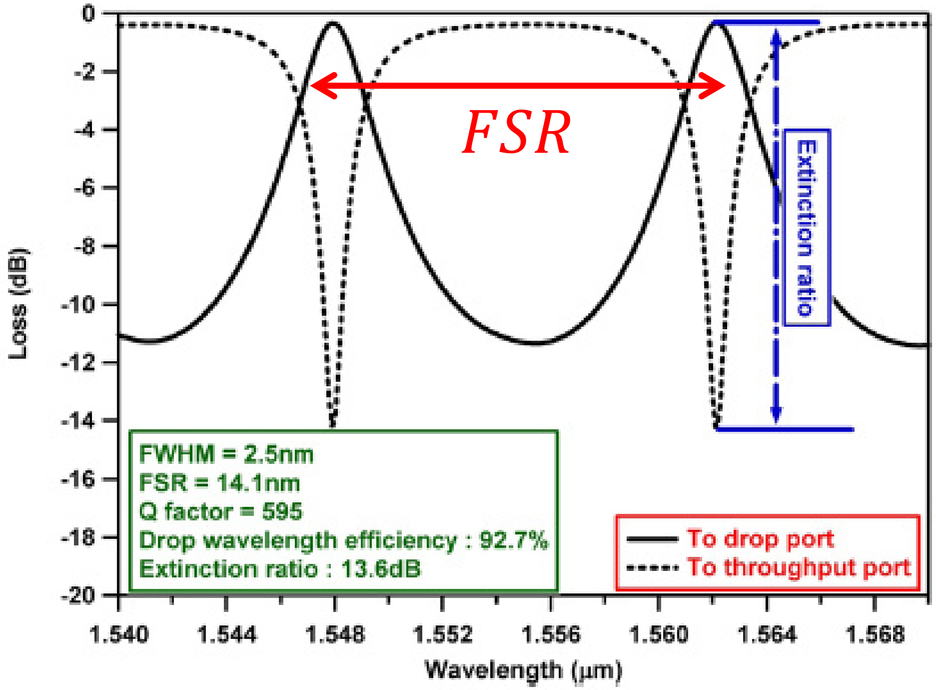
Implementation



Optical Ring Resonators

Take advantage of resonance





Resonance condition: $n_w L = N\lambda$

$$FSR = \frac{c}{n_w L}$$

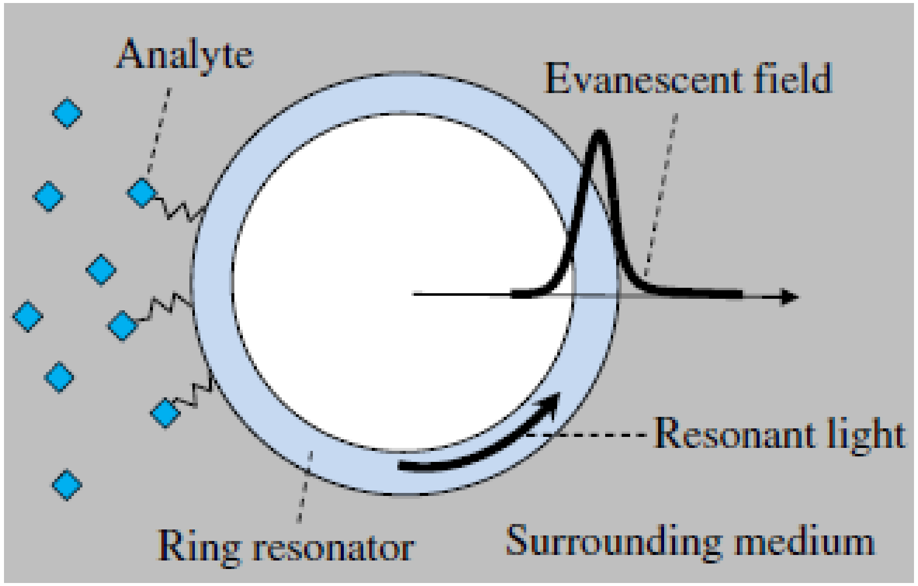
Q depends on loss and coupling

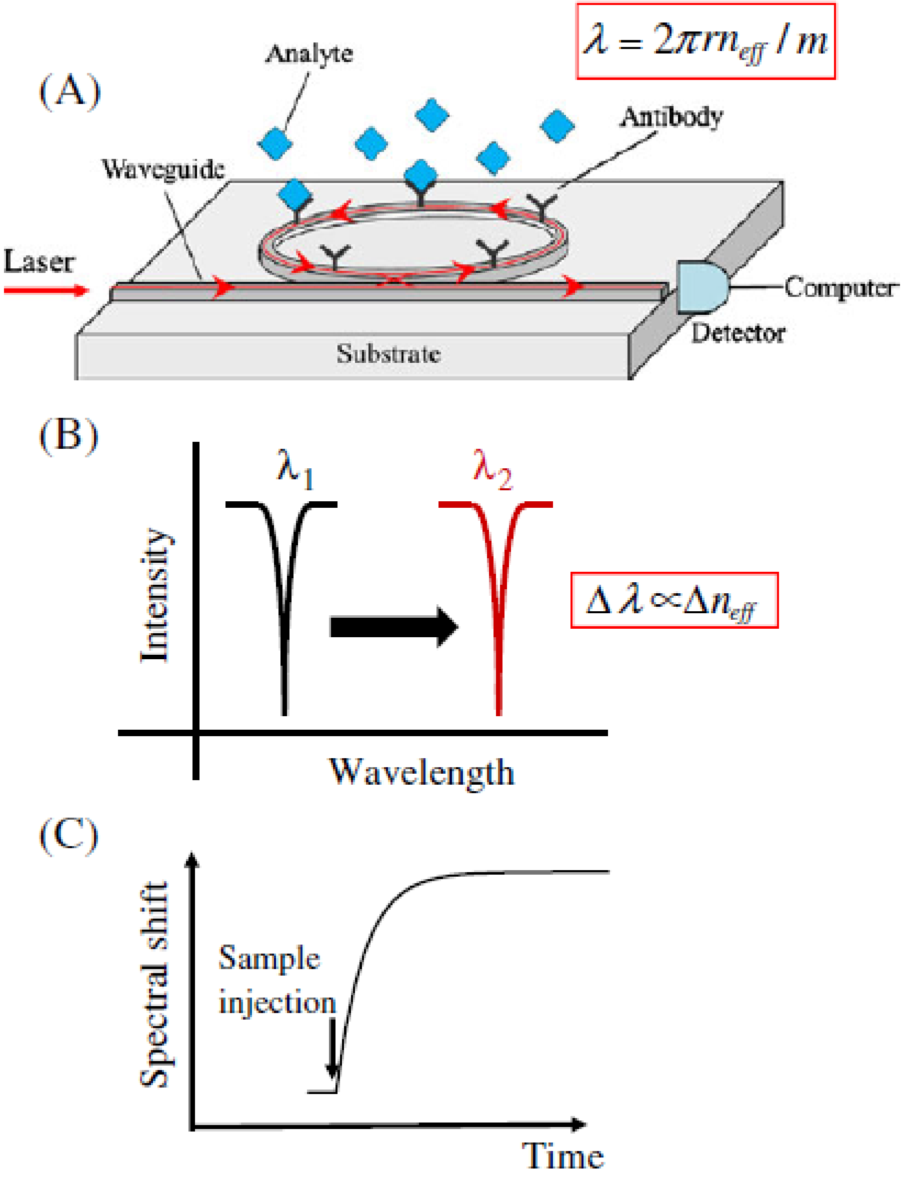
Free Spectral Range

Biosensors Ring Resonators

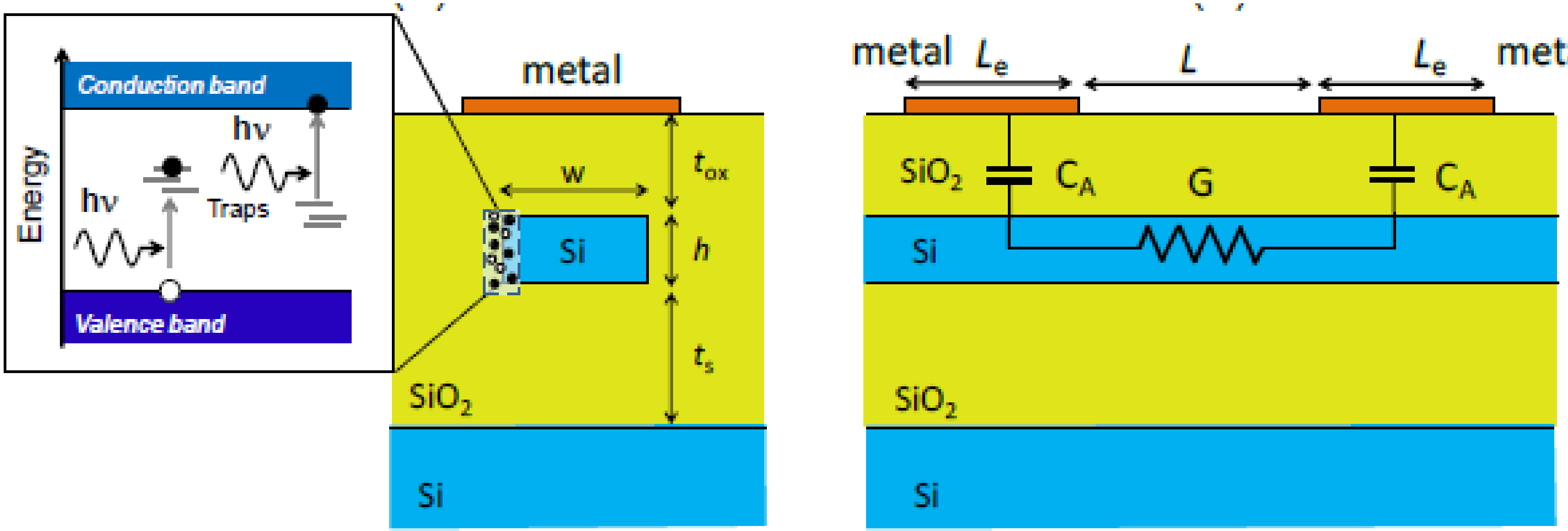
The presence of the analyte changes *n*

Change in the resonance frequency





A Transparent In-Line Light Power Monitor



Invented at Polimi!

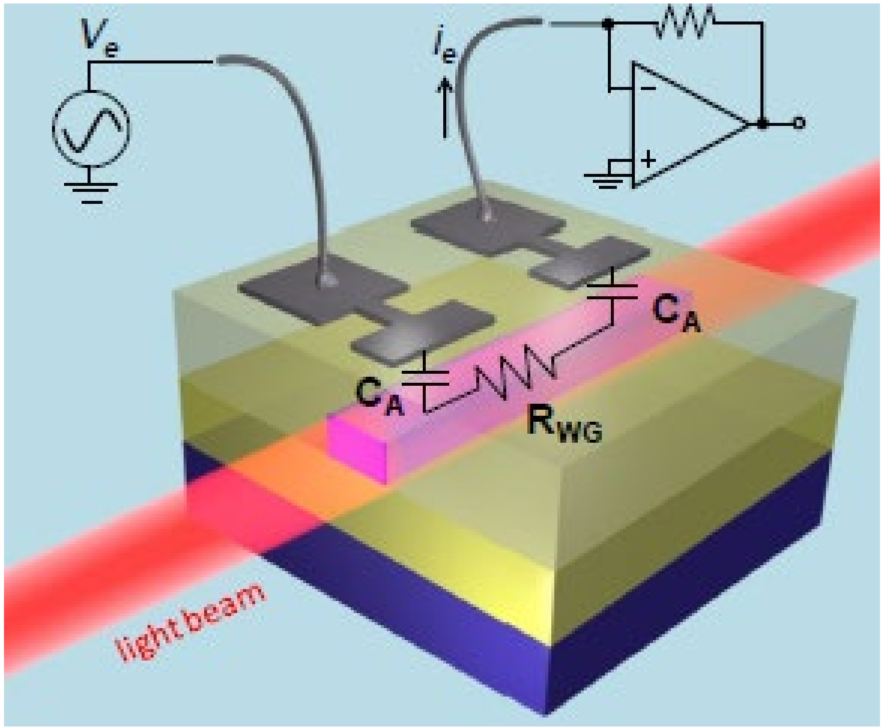
$$\lambda[\mu m] = \frac{1.24}{E_{gap}[eV]}$$

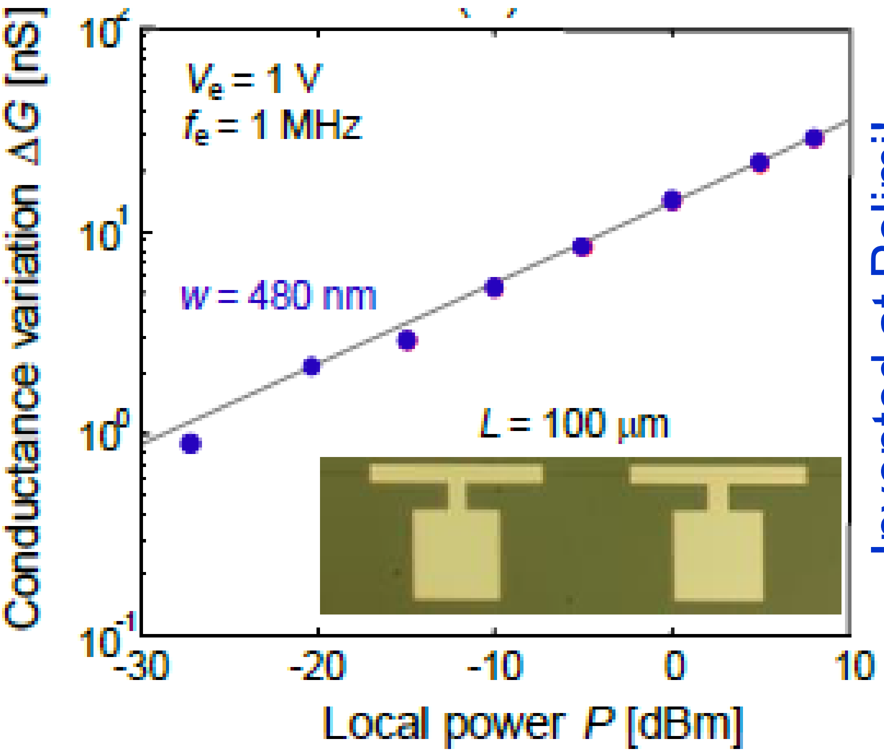
$$C_A = \varepsilon_{SiO2}\frac{L_e w}{t_{ox}} \qquad G = \sigma_{Si}\frac{hw}{L_e}$$

Although Si is transparent at 1550nm, the defects at the surface produce inter-gap energy states that allow photon absorption

Insulated far top metal electrodes do not perturb the optical field

A Transparent In-Line Light Power Monitor





Invented at Polimi!

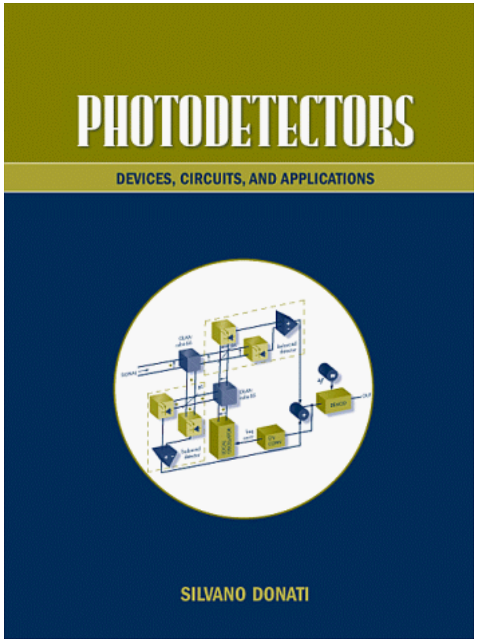
Local power intensity can be monitored by a contactless AC measurement of the conductivity of the Si waveguide

$$f_e > \frac{1}{2\pi C_A / 2R_{WG}} \qquad R_{WG} = \frac{1}{G}$$

Key Take-Home Messages

- Fluorescent microscopy leverages fluorescent molecular markers (first GFP) that can be genetically manipulated.
- In a biochip the optical system comprises: a source (LED, laser, lamp) and a photodetector.
- To map **surfaces** below the optical diffraction limit, scanning probe techniques (AFM in particular) can be used.
- Integrated photonic waveguides (arranged in interferometer or resonators devices) whose cladding is functionalized with receptors can be used to detect molecular binding. The quantitative relation between the analyte concentration and Δn is critical and need to be elucidated for each sensor.

References



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Non-Invasive On-Chip Light Observation by Contactless Waveguide Conductivity Monitoring

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